



SURFACE VEHICLE RECOMMENDED PRACTICE

J1939™-71

FEB2025

Issued 1994-08
Revised 2025-02

Superseding J1939-71 AUG2022

(R) Vehicle Application Layer

RATIONALE

This document has been reorganized for clarification and better understanding. This revision has various corrections and references for ASCII and text file usage to current standards, and a new statement of usage for "Signal" and "Parameter" was added to enhance consistency going forward when adding new PGN and SPN definitions.

FOREWORD

The SAE J1939 communications network is defined using a collection of individual SAE J1939 documents based upon the layers of the Open System Interconnect (OSI) model for computer communications architecture. The SAE J1939-71 document defines the majority of the OSI application layer data parameters (SPs) and messages (PGs) that are relevant to most SAE J1939 applications. This document also defines the conventions and notations for data encoding and parameter placement in PG data.

The SAE J1939 communications network is a high-speed ISO 11898-1 CAN-based communications network that supports real-time closed loop control functions, simple information exchanges, and diagnostic data exchanges between electronic control units (ECUs) physically distributed throughout the vehicle.

The physical layer aspects of SAE J1939 reflect its design goal for use in heavy-duty environments. Horizontally integrated vehicles involve the integration of different combinations of loose package components, such as engines and transmissions that are sourced from many different component suppliers. The SAE J1939 common communication architecture strives to offer an open interconnect system that allows the ECUs associated with different component manufacturers to communicate with each other.

SAE Executive Standards Committee Rules provide that: "This report is published by SAE to advance the state of technical and engineering sciences. The use of this report is entirely voluntary, and its applicability and suitability for any particular use, including any patent infringement arising therefrom, is the sole responsibility of the user."

SAE reviews each technical report at least every five years at which time it may be revised, reaffirmed, stabilized, or cancelled. SAE invites your written comments and suggestions.

Copyright © 2025 SAE International

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, transmitted, in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, or used for text and data mining, AI training, or similar technologies, without the prior written permission of SAE.

TO PLACE A DOCUMENT ORDER: Tel: 877-606-7323 (inside USA and Canada)
Tel: +1 724-776-4970 (outside USA)
Fax: 724-776-0790
Email: CustomerService@sae.org
http://www.sae.org

SAE WEB ADDRESS:

For more information on this standard, visit

https://www.sae.org/standards/content/J1939/71_202502/

TABLE OF CONTENTS

1.	SCOPE.....	4
2.	REFERENCES.....	4
2.1	Applicable Documents	4
2.1.1	SAE Publications.....	4
2.1.2	ISO Publications.....	4
3.	DEFINITIONS	5
4.	ABBREVIATIONS	5
5.	TECHNICAL REQUIREMENTS.....	5
5.1	Parameter Group (PG).....	5
5.1.1	Parameter Group Number (PGN)	6
5.1.2	PG Label (PG Name) and Acronym.....	7
5.1.3	PG PDU Type	7
5.1.4	Default Priority.....	8
5.1.5	PG Data Length Specification	8
5.1.6	PG Data and PG Data Definition	9
5.1.7	PG Data Bit Numbering.....	9
5.1.8	SP Position (SP Position in PG)	10
5.1.9	Parameter Data Byte Ordering	10
5.1.10	Unassigned PG Data Bits	11
5.1.11	Unsupported Parameters	11
5.1.12	Transmission Rate	11
5.2	Suspect Parameter (SP)	14
5.2.1	Suspect Parameter Number (SPN).....	15
5.2.2	Suspect Parameter Classifications	15
5.2.3	SP Label (SP Name).....	15
5.2.4	SP Description	15
5.2.5	SP Type	15
5.2.6	SP Data Definition.....	16
5.2.7	Operational Range	16
5.2.8	SP Data Validity Indicators	16
5.2.9	Parameter Ranges	17
5.3	SLOT	19
5.3.1	SLOT Name	19
5.3.2	SLOT Identifier	19
5.3.3	SLOT Type.....	19
5.3.4	Transfer Function (Data Encoding).....	19
5.3.5	Scaling (Scale Factor).....	20
5.3.6	Offset.....	20
5.3.7	Length (Data Length)	20
5.3.8	Unit.....	21
5.3.9	Range (Data Range).....	21
5.4	Text Data (ASCII) Parameters	21
5.4.1	Text Data Characters	21
5.4.2	Text Data Byte Order	22
5.4.3	Fixed Length Text SLOT Model	22
5.4.4	Character Delimited, Variable Length Text SLOT Model.....	23
5.4.5	Byte Count Delimited, Variable Length Text SLOT Model.....	25
5.5	SP Position Notation and PG Data Placement	26
5.5.1	SP Position Notation Classifications for Data Placement	26
5.5.2	Parameter Data Length Classifications for Data Placement.....	26
5.5.3	Parameter Data Placement Illustrations	27
5.5.4	Integer Byte Length Parameters - SP Position Notation and Data Placement.....	27
5.5.5	Fractional Byte Length Parameters - SP Position Notation and Data Placement	30
5.5.6	Variable Length Parameters - SP Position Notation and Data Placement	32

5.6	Unspecified Bits in the PG Data Definition.....	33
5.6.1	Unspecified Bits - Illustrated Example	34
5.7	General Guidelines	34
5.7.1	Data Acquisition to Data Transmission Timing	34
5.7.2	Parameters with Multiple Sources	34
5.7.3	PG Data Length Variation	35
5.7.4	Retransmitting PG and SP Data	35
5.7.5	Assignment of SLOT for New Parameter (SP)	35
5.7.6	Adding a Parameter to a Parameter Group	35
5.7.7	Naming Convention for Engine Parameters	35
5.7.8	Signal and Parameter Terms	36
6.	NOTES	36
6.1	Revision Indicator.....	36
APPENDIX A	SLOTS.....	37
APPENDIX B	SUSPECT PARAMETERS	38
APPENDIX C	PARAMETER GROUPS	39
APPENDIX D	SUPPORTING INFORMATION	40
Figure 1	Relationship of PGN to EDP, DP, PF, and PS	6
Figure 2	Equations for calculating EDP from PGN	6
Figure 3	Equations for calculating DP from PGN	6
Figure 4	Equations for calculating PF from PGN	7
Figure 5	Equations for calculating PS from PGN	7
Figure 6	Limits of hybrid update rates	12
Figure 7	On-change implementation option 1	13
Figure 8	On-change implementation option 2	14
Figure 9	ISO 8859-1 Latin 1 character set	22
Figure 10	Data placement for single byte data (all types of data).....	29
Figure 11	Data placement for multiple byte data (not text or byte array data).....	29
Figure 12	Data placement for multiple byte data (not text or byte array data).....	29
Figure 13	Data placement for multiple byte data (text or byte array data).....	29
Figure 14	Data placement for fractional byte data (less than 1 byte) within byte boundary	31
Figure 15	Data placement for fractional byte data (less than 1 byte) across byte boundary.....	31
Figure 16	Data placement for fractional byte data (larger than 1 byte) ending on byte boundary.....	32
Figure 17	Data placement for fractional byte data (larger than 1 byte) starting on byte boundary.....	32
Figure 18	Data placement for fractional byte data (larger than 1 byte) starting on byte boundary.....	32
Figure 19	Unspecified bits example	34
Figure 20	Cylinder numbering per SAE J824.....	36
Table 1	PG data length class descriptions.....	8
Table 2	Logical signal ranges	18
Table 3	Transmitted values for 2-bit discrete parameters (measured).....	18
Table 4	Transmitted values for 2-bit control commands (status).....	18
Table 5	Summary of text SLOT models	21
Table 6	SP position notation classifications.....	26
Table 7	Parameter length classifications for SP position notation.....	27
Table 8	Diagrams in SP position notation illustrations.....	27
Table 9	SP position notation for integer byte length parameters (not text or byte array data)	28
Table 10	SP position notation for integer byte length parameters (text or byte array data)	28
Table 11	SP position notation for fractional byte length parameters	31
Table 12	SP position notation for variable length parameters	33

1. SCOPE

The SAE J1939 communications network is developed for use in heavy-duty environments and is suitable for horizontally integrated vehicle industries. The SAE J1939 communications network is applicable for light-duty, medium-duty, and heavy-duty vehicles used on-road or off-road, and for appropriate stationary applications which use vehicle-derived components (e.g., generator sets). Vehicles of interest include, but are not limited to, on-highway and off-highway trucks and their trailers, construction equipment, and agricultural equipment and implements.

SAE J1939-71 is the SAE J1939 reference document describing SAE J1939 parameter (SP) and message (PG) definitions, SLOT (standard data encoding) definitions, conventions and notations used to specify the parameter (SP) placement in PG data, conventions for text data parameters, and conventions for PG transmission rates. This document previously contained the majority of the SAE J1939 OSI application layer data parameters and messages for information exchange between the ECU applications connected to the SAE J1939 communications network. It also contained reference figures and reference information. The data parameters (SPs), messages (PGs), reference figures, and information previously published within this document are now published in SAE J1939DA.

There are several SAE J1939-7X documents that collectively define all of the SAE J1939 application layer data parameters and messages. Diagnostic services and some industry-specific data parameters and messages are documented within other SAE J1939-7X application layer documents. An ECU may simultaneously use and support data parameters and messages from multiple SAE J1939-7X application layer documents.

2. REFERENCES

2.1 Applicable Documents

The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the latest issue of SAE publications shall apply.

2.1.1 SAE Publications

Available from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, Tel: 877-606-7323 (inside USA and Canada) or +1 724-776-4970 (outside USA), www.sae.org.

SAE J824	Engine Rotation and Cylinder Numbering
SAE J1939	Serial Control and Communications Heavy-Duty Vehicle Network - Top-Level Document
SAE J1939-21	Data Link Layer
SAE J1939-22	CAN FD Data Link Layer
SAE J1939-73	Application Layer - Diagnostics
SAE J1939-81	Network Management
SAE J1939DA	J1939 Digital Annex

2.1.2 ISO Publications

Copies of these documents are available online at <https://webstore.ansi.org/>.

ISO 6429	Information technology - Control functions for coded character sets
ISO 8859-1	Information processing - 8-bit single-byte coded graphic character sets - Part 1: Latin alphabet No. 1
ISO 11898-1	Road vehicles - Controller area network (CAN) - Part 1: Data link layer and physical coding sublayer

3. DEFINITIONS

Refer to SAE J1939 (top-level document) for terms and definitions that are not defined in this document.

3.1 BYTE ARRAY

A sequence of one or more whole bytes of binary data treated as an array of bytes. See 5.3.4.5.

3.2 FLOOR

Mathematical operation that returns the nearest integer rounded toward zero, i.e., the whole part with the decimal part truncated.

4. ABBREVIATIONS

Refer to SAE J1939 (top-level document) for additional abbreviations that are used in this document.

>> Bitwise logical right shift operator

% Mathematical modulo operator

5. TECHNICAL REQUIREMENTS

The application layer provides a means for application processes to access the OSI environment. This layer contains management functions and generally useful mechanisms to support applications.

5.1 Parameter Group (PG)

An SAE J1939 Parameter Group, or SAE J1939 message, is a standardized data object (PG Data) that contains data for a predefined group of zero or more data parameters. All data and information communicated over SAE J1939 is done using SAE J1939 PGs. The SAE J1939 Parameter Group Number (PGN) is the unique numeric identifier assigned to each SAE J1939 Parameter Group. The technical definition for an SAE J1939 PG specifies its identity characteristics (PGN, PG Name, Acronym), its PG data definition (data content, PG data length variability), and its transmit characteristics (transmission rate, addressing capability); see 5.1.1 through 5.1.12.

This document only describes the properties of PGs with a PGN value on PGN Page 0 and PGN Page 1.

Most SAE J1939 PGs are fully specified in SAE J1939DA or one of the other SAE J1939 documents. Some SAE J1939 PGs are assigned to other industry standards, and the definitions for those SAE J1939 Parameter Groups are maintained within the respective industry standard documents.

Within SAE J1939 documents, a reference to a specific PG is commonly done using either the PG Name or the PG Acronym, followed by the PGN in parenthesis. For example, Diagnostic Message 1 PG (PGN 65225) or DM1 (PGN 65225).

All transmitted SAE J1939 PGs on an SAE J1939 network shall only be standardized SAE J1939 PG and shall conform to the SAE J1939 specified characteristics including, but not limited to, transmission rate, data content, data placement, and data encoding. The PG data for each transmitted SAE J1939 PG shall only contain data for the parameters specified for that SAE J1939 PG and encoded as specified for each parameter. The only exception is the data communicated using SAE J1939 PG defined as proprietary messages, i.e., PropA, PropA1, PropB, and PropB1 PGs, where the parameter data and its placement in the PG data is defined by the transmitter of the proprietary messages.

An SAE J1939 PG describes a PDU above the data link layer; therefore, the length of the PG data for a PG can be longer than the maximum length of the data field of a single CAN data frame, e.g., 8 bytes for a CEFF data frame or 64 bytes for an FEFF data frame. When the PG data for a PG cannot be sent in its entirety in the data field of a single SAE J1939 data frame, then an SAE J1939 transport protocol is used to transfer the PG data using a series of SAE J1939 data frames, as described in SAE J1939-21 and SAE J1939-22 (SAE J1939 data link layers).

5.1.1 Parameter Group Number (PGN)

The Parameter Group Number (PGN) is an 18-bit numeric value that uniquely identifies an SAE J1939 PG. The PGN is most commonly used in SAE J1939 communications to identify the PG associated to the PG data being communicated. The PGN can be communicated as part of the PG Data to identify or reference a specific PG. The PGN can be cited in documentation, specifications, and dialog to refer to a specific PG. Due to the SAE J1939 standard for PGNs assigned to PDU1 type PGs, there are only 8672 valid PGNs out of the 131072 possible PGNs on PGN Page 0 and PGN Page 1.

For PGN Page 0 and PGN Page 1 PGNs, there are four individual component values derived from the numeric value: Extended Data Page (EDP), Data Page (DP), PDU Format (PF), and PDU Specific (PS). The relationship between the PGN value and these four component values is according to the equation in Figure 1 for PGNs on PGN Page 0 and PGN Page 1.

$$\text{PGN} = (\text{EDP} * 2^{17}) + (\text{DP} * 2^{16}) + (\text{PF} * 2^8) + \text{PS}$$

Figure 1 - Relationship of PGN to EDP, DP, PF, and PS

5.1.1.1 PGN Page

The 262144 possible PGN values are partitioned into four ranges of 65536 values, known as PGN pages. Each PGN Page is enumerated according to the 2-bit numerical value comprised of the most significant 2 bits of the 18-bit PGN value, specifically the EDP and DP values. A Page 0 PGN has an EDP value of 0 and a DP value of 0. A Page 1 PGN has an EDP value of 0 and a DP value of 1.

5.1.1.2 Extended Data Page (EDP)

The Extended Data Page is a 1-bit value of the four component values that comprise the SAE J1939 PGN and one of two component values that identify the Page of the PGN. The EDP value is the result of the PGN value bit shifted right by 17, as shown by the equations in Figure 2.

$$\text{EDP} = \text{PGN} \gg 17$$

$$\text{EDP} = \text{floor}(\text{PGN} / 131072)$$

Figure 2 - Equations for calculating EDP from PGN

5.1.1.3 Data Page (DP)

The Data Page is a 1-bit value of the four component values that comprise the SAE J1939 PGN and one of two component values that identify the Page of the PGN. The DP value is the least significant bit of the result of the PGN value bit shifted right by 16, as shown by the equations in Figure 3.

$$\text{DP} = (\text{PGN} \gg 16) \% 2$$

$$\text{DP} = (\text{floor}(\text{PGN} / 65536)) \text{ modulo } 2$$

Figure 3 - Equations for calculating DP from PGN

5.1.1.4 PDU Format (PF)

The PDU Format is an 8-bit value of the four component values that comprise the SAE J1939 PGN. There is a correlation between the PF value of the PGN and the PDU Type of the PG. A PDU1 Type PG shall have a PGN with a PF value from 0 to 239, inclusively, and a PDU2 Type PG shall have a PGN with a PF value from 240 to 255, inclusively. The PF value is the least significant byte of the result of the PGN value bit shifted right by 8, as shown by the equations in Figure 4.

$$PF = (PGN \gg 8) \% 256$$

$$PF = (\text{floor}(PGN / 256)) \text{ modulo } 256$$

Figure 4 - Equations for calculating PF from PGN

5.1.1.5 PDU Specific (PS)

The PDU Specific is an 8-bit value of the four component values that comprise the SAE J1939 PGN. The PS value for the PGN of a PDU1 Type PG shall always be 0, and the PS value for a PDU2 Type PG PGN shall be from 0 to 255, inclusively. The PS value is the least significant byte of the PGN value, as shown by the equations in Figure 5.

$$PS = PGN \% 256$$

$$PS = PGN \text{ modulo } 256$$

Figure 5 - Equations for calculating PS from PGN

5.1.2 PG Label (PG Name) and Acronym

A unique PG Name and unique Acronym are assigned to each Parameter Group to aid in referring to the PG in documentation, specifications, and dialog. The PG Name is a title meant only to provide a general description of the parameter data content, e.g., Transmission Control 1, or the function of the PG, e.g., Request. The Acronym for a PG is generally derived from the PG Name, e.g., TC1 for the Transmission Control 1.

5.1.3 PG PDU Type

The PDU Type property describes the destination addressing capability of the PG. There are two PDU types assigned to SAE J1939 PGs: PDU1 or PDU2.

A PDU1 type PG is commonly referred to as a destination specific PG. Each instance of a PDU1 type PG can be addressed (sent) to a different controller application (CA) (e.g., network device). A PDU1 type PG can be addressed to a single CA or globally to all CAs (using the Global Destination Address). PDU1 type is best suited for a PG with control or command information in its PG data, where it can be necessary to send the PG with different parameter data values to different CAs. For example, sending actuator position commands to different actuators. Some PDU1 type PGs have special destination address requirements. For example, the Address Claimed PG and the Acknowledgement PG, both PDU1 type PGs, each require using the global address (255) as the destination address.

A PDU2 type PG is commonly referred to as a broadcast PG and does not support destination addressing. A PDU2 type PG can only be sent globally to all devices on the network. A PDU2 type PG cannot be directed only to a specific receiver. A PDU2 type is best suited for a PG with measured or status data, such as engine speed or oil temperature, where the reported PG data values are the same regardless of the receiver.

There is a correlation between the PDU Type of the PG and the PDU Format (PF) value of the assigned PGN. The PDU Format value for a PDU1 PG is from 0 to 239, inclusively, and the PDU Format value for a PDU2 PG is from 240 to 255, inclusively.

5.1.4 Default Priority

The Default Priority for a PG specifies the recommended data link layer (D_PDU) priority to use when transmitting an instance of the PG. Refer to SAE J1939-21 and SAE J1939-22 for more details.

5.1.5 PG Data Length Specification

PG Data Length is a specification of the length of the PG data for the PG. The PG data for a PG can be defined as having a fixed (non-variable) length or having a variable length. For a PG defined with fixed length PG data, the length of the PG data is the specified byte length for every instance of that PG. For a PG defined with variable length PG Data, the length of the PG data can vary between instances for that PG. A PG is defined with variable length PG data when its PG data definition contains parameters with variable length data, contains repeating groups of parameters, and/or its PG data definition can be modified and extended to accommodate additional parameters.

SAE J1939 uses numerous methods for specifying the length of the PG data. In SAE J1939 and some SAE J1939 documents, the length characteristic of the PG data is defined using only the “PG Data Length” property. In SAE J1939DA and other SAE J1939 documents, the length characteristic of the PG data is defined by the combination of “PG Data Length Class,” “PG Data Minimum Length,” and “PG Data Length Constraint” properties.

5.1.5.1 PG Data Length

The PG Data Length property specifies the length characteristic of the PG data. When the PG Data Length specifies a single numeric value, the length of the PG data for all instances of the PG should be that specified length. If the PG Data Length specifies “variable” or a range of numeric values, then the length of the PG data can vary for the PG.

SAE J1939 is transitioning from using the PG Data Length property to using the combination of “PG Data Length Class,” “PG Data Minimum Length,” and “PG Data Length Constraint” properties for specifying the length characteristics of the PG data. The transition allows SAE J1939 to better distinguish when a PG has fixed length PG Data that shall never change, and a PG has fixed length PG Data for all instances by an originator, but the length of the PD Data can change over time as new SPs are added to the PG data definition.

5.1.5.2 PG Data Length Class

PG Data Length Class property describes the variability in the length of PG Data per PG data definition. It also describes the ability for the PG data definition to be lengthened to add more data parameters. Each of Data Length classes are described in Table 1. The PG Data Length Class property is used together with the PG Data Minimum Length and PG Data Length Constraint properties.

Table 1 - PG data length class descriptions

Classification	Description
Fixed	PG Data length shall always be <i>PG Data Minimum Length</i> for all instances of PG. The PG data definition shall not be extended longer than <i>PG Data Minimum Length</i> . The PG data definition only has single instances of fixed length parameters (SPs).
Fixed Extendable	PG Data length can be as short as <i>PG Data Minimum Length</i> , can vary between instances from different senders, and should be constant between instances from same sender. The PG data definition can be extended (by the SAE J1939 Committee) with new bytes beyond last defined byte to accommodate new parameters (SPs). The PG data definition only has single instances of fixed length parameters (SPs).
Variable	PG Data length can be as short as <i>PG Data Minimum Length</i> and can vary between all instances of the PG. The PG data definition can be extended (by the SAE J1939 Committee) with new bytes beyond last defined byte to accommodate new parameters (SPs). The PG data definition contains variable length SPs and/or repeated groups of SPs.

5.1.5.3 PG Data Minimum Length

PG Data Minimum Length property specifies the shortest permissible length of the PG Data for any instance of the PG. This does not mandate nor require support of the parameters whose data is positioned in the bytes within the specified minimum length. The value of this property is established at PG assignment and is never changed. Bytes associated with unassigned data bits and unsupported parameters are encoded as specified in 5.1.10 and 5.1.11.

For one example, consider a Fixed Extendable PG specified with a PG Data Minimum Length of 15 bytes where a transmitter of the PG supports SPs whose data occupy PG data through byte 11. In this situation, the transmitter shall append 4 bytes of FFh following the 11 bytes spanning the supported SPs so the transmitter sends the PG with PG data that is at least 15 bytes long, i.e., the specified PG Data Minimum Length for the PG.

For another example, consider a Fixed Extendable PG specified with a PG Data Minimum Length of 15 bytes and defined with SPs whose data occupy PG data through byte 22. In this example, a transmitter of the PG supports SPs whose data occupy PG data through byte 18. In this situation, the transmitter is not required to append any bytes of FFh following the 18 bytes spanning the supported SPs since the length of the PG data is equal to or greater than the specified PG Data Minimum Length for the PG. The transmitter in this example is permitted to send the PG with PG data that is 18 bytes long.

5.1.5.4 PG Data Length Constraint

PG Data Length Constraint property describes any special constraint on the length of PG Data. Most commonly used when there is a need to limit the maximum length of the PG data.

5.1.6 PG Data and PG Data Definition

PG Data refers to the data object containing the data communicated by the PG. The data content and structure of the PG Data for a PG is specified by its PG Data Definition and PG Data length properties.

PG Data Definition refers to the specification of each parameter whose data is communicated in the PG Data and the position of the parameter's data within that structured data object. For each data parameter (SP) communicated within the PG data, the PG data definition references the data parameter (SP) by its assigned SPN and specifies the position of the parameter's encoded data in the PG data structure. The SP definition specifies the technical details such as parameter name, data size, and data encoding, as discussed in 5.2. The PG data position for the data for a parameter is specified as either an absolute position (e.g., bytes 4 to 5) or a relative position (subsequent placement following data for another parameter), as discussed in 5.1.8.

The PG data for some PGs can contain multiple instances of data for a single SP, multiple instances of data for a group of SPs, or both. When a PG can contain multiple instances of SPs in the PG Data, the SP or group of SPs are specified in the PG data definition and a description of the repeated data is described in the PG Description. Some examples are Engine Maximum Net Torque Map PG (PGN 64248) and Active Diagnostic Trouble Codes (DM1) PG (PGN 65226).

SAE J1939 specifies a technique for encoding PG data bits for unassigned (undefined) bits and bits for unsupported data parameters, as discussed in 5.1.10 and 5.1.11. This technique permits unassigned bits to be assigned to parameter data in the future without causing incompatibilities.

Parsing parameter data from PG data requires knowing the position of parameters per the PG data definition for that PG and knowing the data parameter's (SP) data definition for the data size and its encoding. For data parameters with relative position, parsing the parameter data can require evaluating or parsing other data parameters.

5.1.7 PG Data Bit Numbering

SAE J1939 uses a one-based index numbering when specifying byte and bit positions in the PG data. Using a Big-endian model, byte 1 is the leftmost byte, byte 2 follows to the right, and so on. Bit 1 is the lowest order bit in a byte, and bit 8 is the highest order bit in a byte. Parameter position notations are described in detail in 5.5.

5.1.8 SP Position (SP Position in PG)

The SP Position specifies the placement of the data for the SP in the PG data. See 5.5 for details on conventions for specifying SP Position in PG and the placement of the SP data.

The following guidelines provide the basis for the parameter data placement conventions. These guidelines and the conventional parameter placement methods should be applied when defining the placement of parameters in PGs.

- a. Parameters with less than 8 bits should reside within a byte boundary.
- b. Parameters with more than 8 bits should either start or stop on a whole byte boundary.
- c. Only parameters with more than 8 bits should span a byte boundary.
- d. Text data parameters, variable length parameters, and parameters with repeating data should start and stop on whole byte boundaries.
- e. Byte ordering rules are specified in 5.1.9.

5.1.9 Parameter Data Byte Ordering

This section specifies the byte ordering of parameter data in the PG data. This applies to all parameters with more than 1 byte of data. Data bits within a byte boundary are always placed (ordered) so the most significant bit of the data is placed in the highest order bit occupied by the parameter data and the least significant bit of the data is placed in the lowest order bit occupied by the parameter data.

Further description of bit placement within PG data is described in 5.5.

5.1.9.1 Default Data Ordering

Parameters with more than 1 byte of data, other than text or byte array data, shall be placed (ordered) least significant byte first in the PG data. Least significant byte (LSB) first for data with integral number of bytes means the least significant byte of the encoded data is positioned in the lowest ordered byte of the PG data bytes occupied by the parameter data, and the most significant byte of the encoded data is positioned in the highest ordered byte of the PG data bytes occupied by the parameter data.

5.1.9.2 Text Data Ordering

Parameter data with text data encoding (see 5.3.4.2), formerly referred to as ASCII, shall be placed (ordered) most significant byte first in the PG data. Most significant byte (MSB) first for text data means the individual characters are positioned in the PG data in left-to-right reading order of the text string. The leftmost character of the text string shall be positioned in the lowest ordered byte of the PG data bytes occupied by the parameter data, and the rightmost character of the text string shall be positioned in the highest ordered byte of the PG data bytes occupied by the parameter data. For example, if the text string is "The quick brown fox jumped over the lazy dog" and the PG definition indicates the parameter data is to start at byte 1 in the PG data, then the text character "T" shall be positioned in byte 1 of the PG data and the text character "g" shall be positioned in byte 44 of the PG data.

5.1.9.3 Byte Array Data Ordering

Parameter data with byte array data encoding (see 5.3.4.5) shall be placed (ordered) in the PG data as follows: the data byte with array index 1 is placed first, followed by the data byte with array index 2, etc. Byte array data refers to a sequence of one or more whole bytes of binary data treated as an array of bytes.

5.1.10 Unassigned PG Data Bits

PG data definitions can have PG data bits that are not yet assigned to any parameter. SAE J1939 refers to these bits as unassigned, unspecified, or SAE J1939 reserved bits. These unassigned bits are strictly reserved for future assignment by SAE J1939 and cannot be used at the discretion of component and system designers.

The “SAE J1939 not available bit encoding” technique (described in 5.1.11) is used to encode PG data bits not yet assigned to a data parameter. This unused PG data bit encoding practice allows SAE J1939 to assign additional data parameters to the unused bits at some future date in order to maximize the data content of the PG data. This encoding technique allows a recipient of the PG data to determine if data is provided for data parameters assigned to those PG data bits even when a data parameter is assigned to those PG data bits in the future.

5.1.11 Unsupported Parameters

The transmitter of an SAE J1939 PG is not required to provide actual parameter data for every SP that is contained in the PG data. The “SAE J1939 not available bit encoding” technique is used to encode PG data bits associated with an unsupported data parameter.

SAE J1939 employs the “SAE J1939 not available bit encoding” technique for encoding PG data bits that are not supported by the transmitter or unassigned data bits. Per this data encoding practice, each PG data bit is encoded with a numerical value of one (1). This practice allows a recipient of the PG data to determine if data is provided for any data parameter assigned to those PG data bits. In cases where “Measured” and “Control” parameters use 2-bit data (11b), these are used as “not available” or “don’t care.”

5.1.12 Transmission Rate

The transmission rate specifies the required transmit behavior and timing requirements for the PG. Most SAE J1939 PGs are transmitted in either a continuous unsolicited behavior or a solicited (requested) behavior. While there are numerous transmission rate models, the most common transmission rate models are continuous unsolicited transmit (e.g., fixed interval period and fixed interval period with event-driven updates) and solicited transmit (e.g., on request). For a PG with continuous unsolicited transmit behavior, the PG is sent automatically and periodically by a transmitter (producer) of the PG. For a PG with solicited transmit behavior, the PG is only sent when requested or prompted. The Request PG (PGN 59904; refer to SAE J1939-21 or SAE J1939-22) provides the service for requesting (soliciting) another device to transmit a specific PG. The PG data for the Request PG contains the PGN of the PG being requested to be transmitted.

All Parameter Group transmission repetition rates defined in SAE J1939 are nominal rates. The actual transmission repetition rate on the network should be at this rate plus/minus the “typical” jitter which occurs in microcontroller-based systems. The average rate should be the nominal value.

5.1.12.1 Transmission Rate for Engine Speed and Crank Angle Based Update Behavior

The transmission rate specified for some PGs include the declaration “engine speed dependent.” This clause can be specified in the PG transmission rate for a PG when the PG data contains parameters that can be calculated, measured, and/or updated based on engine crank angle rather than at a fixed time interval. The primary goal with this allowance is to minimize the latency associated with sampling, calculating, and transmitting the data without overburdening the network, such as when the engine is not running or running at idle speeds. This allowance provides the producer the ability to change the interval between successive instances of the PG based on the current operating speed, keeping the intervals within an acceptable range (see Figure 6).

There are many approaches to sampling the data to be converted and sent over the network. The two preferred approaches are: (a) time-based sampling, calculating, and transmit; and (b) a hybrid of time-based and engine crank angle-based sampling, calculating, and transmit, where timing is modified based on the current operating speed in order to maintain an update rate within an acceptable range (see Figure 6). Because there are multiple ways to acquire and transmit data onto the network, the following guidelines have been defined for the engine speed and directly associated data.

- a. At speeds above 500 rpm, the time from sampling to message transmission shall not exceed 12 ms. Systems that acquire engine speed dependent information via period measurement inherently have less time delay at higher speeds.
- b. Above 1000 rpm, for instance, the time from sampling to message transmission shall range from 5 to 30 ms. Less time is required because the period measurement takes less time at higher speeds. How much time is saved depends on the number of crank angle degrees used to perform the period measurement.

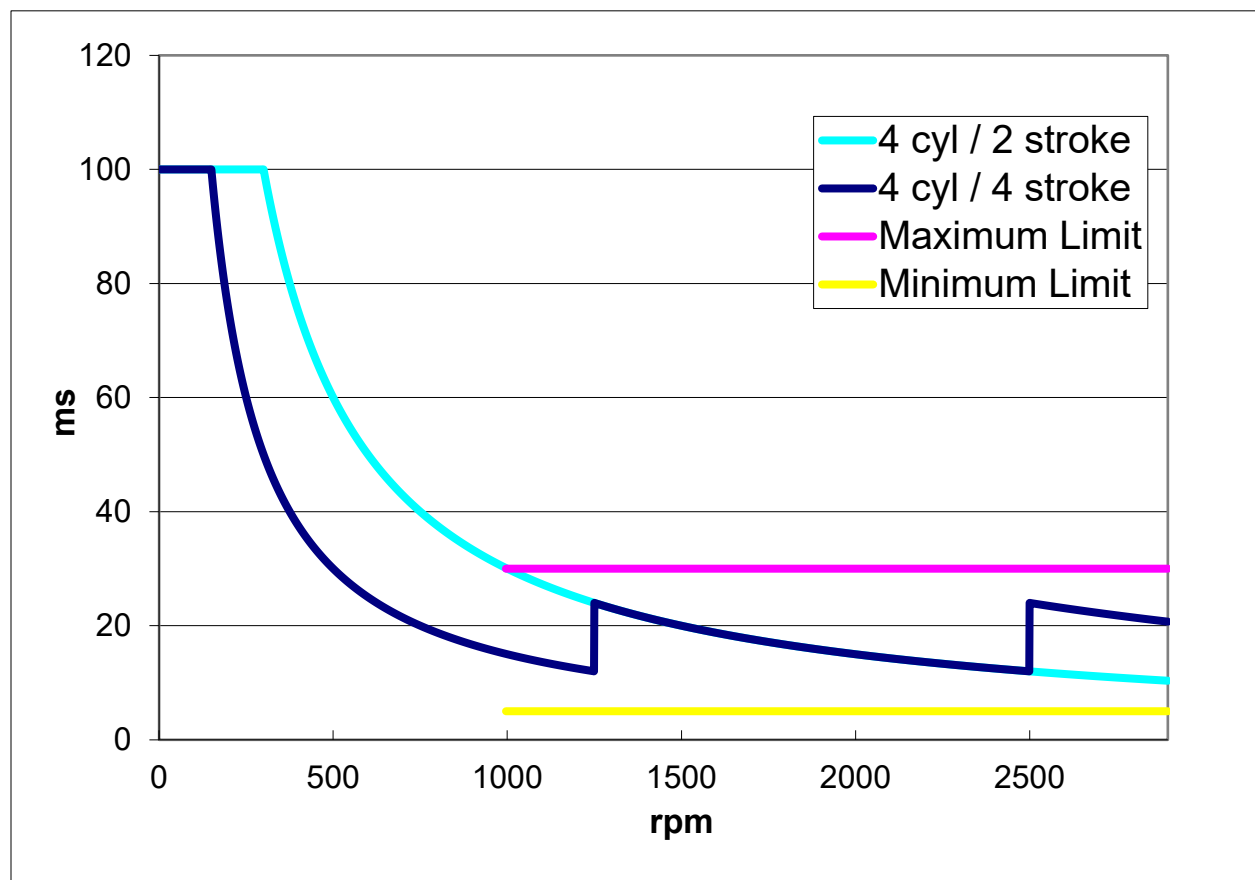


Figure 6 - Limits of hybrid update rates

5.1.12.2 Transmission Rate with On-Change Update Behavior

The transmission rate specified for some PGs specify a fixed periodic transmit interval with an allowance for transmitting additional and immediate instances of the PG when the data changes for one or more data parameters. For example, it can be desirable to immediately transmit the Engine Configuration PG when the torque map changes rather than waiting for the next periodic update interval. An on-change transmission rate allowance specifies a minimum interval between successive instances of the PG to address situations where parameter data can change values at a very high rate and would otherwise trigger excessive instances of the PG and unnecessarily increases bus loading.

For such Parameter Groups, the PGs are defined with a transmission rate in the form of:

Every *MAXUPDATEPERIOD* and on *CHANGECRITERIA*, but no faster than every *MINUPDATEPERIOD*

where:

CHANGECRITERIA = The data change criterion that prompts an immediate transmit of a new instance of the PG.

MAXUPDATEPERIOD = The maximum period between successive instances of the PG. When *CHANGECRITERIA* is not satisfied, this is the preferred period of the PG.

MINUPDATEPERIOD = The minimum period between any two successive instances of the PG. If a *CHANGECRITERIA* data change occurs within *MINUPDATEPERIOD* since the last transmitted instance, then the transmit of the instance with the *CHANGECRITERIA* data change shall not be transmitted until *MINUPDATEPERIOD* time following the last transmitted instance. This does not apply to the first PG instance after a periodic broadcast.

Two acceptable implementations are illustrated in Figures 7 and 8. In each illustration, the horizontal line represents time; the vertical bars topped with a numbered circle represent message instances; and the diagonal line represents a *MAXUPDATEPERIOD* timer that counts down to zero, which triggers the transmission of a periodic interval message instance. In both illustrations, all shown message instances are triggered by *MAXUPDATEPERIOD*, except for Message 2, which is triggered by *CHANGECRITERIA*.

Figure 7 shows an implementation method where the message instances triggered by *CHANGECRITERIA* do not change the *MAXUPDATEPERIOD* timing between successive periodic message instances. In the illustration of this method, Message 1 represents a periodic instance triggered by the elapse of the *MAXUPDATEPERIOD* time since previous (unshown) periodic instance. Message 2 is triggered by *CHANGECRITERIA* but does not reset the *MAXUPDATEPERIOD* countdown timer. Since the *MAXUPDATEPERIOD* countdown timer is not reset by the *CHANGECRITERIA* instance, Message 3 is transmitted when *MAXUPDATEPERIOD* elapses since Message 1. Message 4 is transmitted when *MAXUPDATEPERIOD* elapses since Message 3.

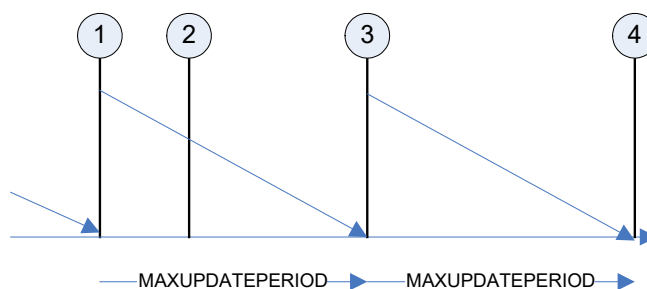


Figure 7 - On-change implementation option 1

Figure 8 shows an implementation method where the *MAXUPDATEPERIOD* interval period is based on the last transmitted message instance. In the illustration of this method, Message 1 represents a periodic instance triggered by the elapse of the *MAXUPDATEPERIOD* time since previous (unshown) periodic instance. Message 2 is triggered by *CHANGECRITERIA* and the *MAXUPDATEPERIOD* timer. Message 3 is triggered by the elapse of the *MAXUPDATEPERIOD* time since Message 2; this implementation results in Message 3 being transmitted at a later time than if *CHANGECRITERIA* had not been satisfied. This implementation can have a lower average bus loading compared to the implementation shown in Figure 7, as illustrated by the absence of Message 4 in Figure 8 over the same overall time shown in both illustrations.

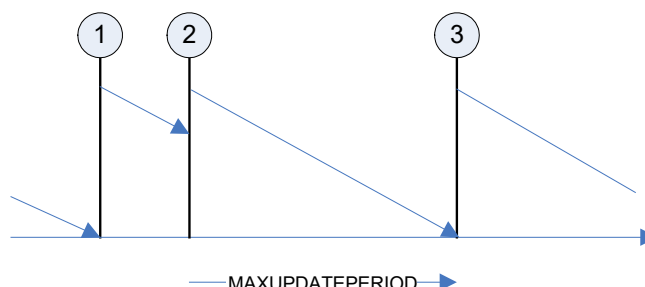


Figure 8 - On-change implementation option 2

The two acceptable implementations, illustrated in Figures 7 and 8, were established after many “on change” PGs were published. As a result, the implementation of some PGs can vary from the implementations described in this section. This section is intended to provide uniformity of future implementations of “on change” messages. After July 2010, new implementations of “on change” PGs are expected to conform to the one of these acceptable implementations. There are implementations done prior to that date that comply with this definition, and no change is required. While this section describes the preferred implementation, existing implementations done prior to July 2010 are grandfathered and can have an alternate acceptable definition.

5.1.12.3 Transmission Rate Deviation When Used in Diagnostic Applications

When a PG is transmitted on a control network where there can be consumers using the PG data for control functions, then the documented standard PG transmission rate shall be used by the transmitter (producer).

However, in an OEM-integrated vehicle system, the vehicle system designer can have an SAE J1939 network where the PG and its data is not required for control functions, and the PG data is only required for non-control (or informational only) functions, such as user interface display or diagnostic tools. In such an OEM-integrated vehicle system, it is permissible for that SAE J1939 network to use an extended (i.e., longer interval time), system-specific transmission rate for the periodic PG. The extended transmission rate can be up to 5X slower (longer) than the defined transmission rate. For example, the extended transmission rate for a PG defined with 50 ms transmission rate can be up to 250 ms. The responsibility falls on the vehicle system designer of the OEM vehicle system to make sure all involved ECUs function appropriately with the system-specific, extended rate.

5.2 Suspect Parameter (SP)

The SAE J1939 Suspect Parameter (SP) refers to a diagnosable data parameter, component, event, or condition. Most SAE J1939 SPs include an SP description that provides an operational definition of the data parameter, component, event, or condition.

Within SAE J1939 documents, a reference to a specific SP is commonly done using either the SPN or the SP Label followed by the SPN in parenthesis. For example, Transmission Requested Gear (SPN 525).

5.2.1 Suspect Parameter Number (SPN)

SAE J1939 assigns a unique numeric value, called the Suspect Parameter Number (SPN), to every defined SAE J1939 SP. The SPN is primarily used as part of an SAE J1939 Diagnostic Trouble Code (DTC) (refer to SAE J1939-73) to identify the system parameter, component, event, or condition that is suspected as having a detected failure or issue. SAE J1939 PG data definitions use the SPN to cross-reference a specific data parameter SP whose data is communicated in the PG data.

The SAE J1939 Committee is the data registration authority for all SPN assignments.

5.2.2 Suspect Parameter Classifications

SAE J1939 SPs are classified as either a diagnostic-only SP or a parametric SP (data parameter SP).

5.2.2.1 Diagnostic Only SP

A diagnostic-only SP is an SAE J1939 SP whose SPN can only be used in diagnostic reporting with an SAE J1939 DTC. A diagnostic-only SP can be changed to data parameter SP (parametric SP) by adding a data definition and associating the parameter to an SAE J1939 PG.

A diagnostic-only SP can be identified as an SP listed in SAE J1939DA with SLOT Name "Diagnostic" (SLOT ID 202).

5.2.2.2 Parametric SP (Data Parameter SP)

A parametric SP is an SAE J1939 SP that has an SP data definition whereby related quantifiable data can be communicated within the PG data of an SAE J1939 PG. The SPN assigned to a parametric SP can also be used in diagnostic reporting with an SAE J1939 DTC, same as a diagnostic only SP.

The SP data definitions for most SAE J1939 parametric SPs are documented in SAE J1939DA or in one of the other SAE J1939 documents, such as SAE J1939-21, SAE J1939-73, and SAE J1939-81. SAE J1939 PG data definitions use the SPN to cross-reference a specific data parameter SP whose data is communicated in the PG data. The SP data definition of the associated parametric SP specifies the encoding used for the parameter data in the PG data. The SP data definition establishes a well-defined interface and strict compliance to the specified characteristics is mandatory.

5.2.3 SP Label (SP Name)

The SP Label provides the unique and official title of the parameter. When there are multiple instances of the same parameter on the same component, the SP Labels for the parameters shall be qualified to differentiate each instance, such as numbering the instances. See 5.7.7 for the convention for naming instances of engine related parameters. The SP Label for an SP can change when new instances of the same parameter are assigned.

5.2.4 SP Description

The SP Description provides an operational definition of the SP data. The SP Description can include the real world meaning of encoded state values, parameter-specific indicators, etc.

5.2.5 SP Type

The SP Type property specifies if the parameter is defined to communicate data based on direct observation (measured) or data not based on direct observation (status), e.g., calculation, action taken, action request. The SP Type property is specified only for data parameter SPs that are defined within SAE J1939 documents.

5.2.5.1 Measured

The SP type “Measured” specifies the parameter data conveys the current value as directly measured or observed by the data producing device. A data producing device shall not report data for a “measured” parameter if the device does not directly measure or observe the data.

Examples of measured data are boost pressure, ignition on/off, cruise set switch activated, maximum cruise speed, cruise set speed, engine speed, percent load at current speed, etc.

5.2.5.2 Status

The SP type “Status” indicates the parameter data conveys a computed value or conveys an action taken or requested by the data producing device. When the data conveys a computed value, the computation can use local and/or network “measured” and/or “status” information. When the data conveys a requested (commanded) action or describes an action taken, measured confirmation of the action is not necessarily assured. For instance, the data can indicate a solenoid has been commanded to be activated, but measurement is not required to ensure the solenoid accomplished this function.

Examples of status data are engine brakes are enabled, PTO speed control is active, cruise control is active, the cruise control is in the “set” state of operation (as opposed to a measured indication that the “set” switch contacts are closed), fault codes, torque/speed control override modes, desired speed/speed limit, engine torque mode, engine’s desired operating speed, engine’s operating speed asymmetry adjustment, etc.

5.2.6 SP Data Definition

The SP data definition of the associated parametric SP specifies the encoding used for the parameter data in the PG data. SAE J1939 specifies the data definition (data encoding) for a parametric SP by reference to a SLOT (Scaling, Length, Offset, Transfer function). Some SAE J1939 SP definitions identify the SLOT by its SLOT Name or SLOT Identifier; other SP definitions, instead, only present the scale factor, data length, offset, and range of the SLOT used for data encoding.

5.2.7 Operational Range

The operational range property for a parametric SP specifies the actual range of engineering values for the parameter. When the data range specified by the SLOT is not appropriate for the parameter, then the operational range specifies the subset of the SLOT data range that is acceptable. If an operational range is blank or omitted for a parametric SP, then the entire data range specified by the SLOT is acceptable for the parameter.

5.2.8 SP Data Validity Indicators

Most SAE J1939 SLOTS allocate some of the possible encoded data values for reporting data validity indicators. Data validity indicators are specific encoded values typically used to indicate that valid data is not provided in the PG data for a parameter. Data validity indicators can be used when the parameter is not supported or when valid parameter data is not available due to some condition, like a detected sensor failure, and are compatible with the SAE J1939 Not Available Data Encoding technique. Because of data validity indicators, the number of encoded data values available for encoding valid parameter data is less than the total number of encoded data values per the data size. For example, a 1-byte SLOT allocate 250 encoded data values for valid parameter data and five values for data validity, whereas a 2-byte SLOT can allocate 64 255 encoded data values for valid parameter data and 1280 values for data validity. The use of data validity indicators means some data parameters need to use a SLOT with a larger data size. For example, a simple two state data parameter that supports data validity indicators needs to use a 2-bit state value SLOT, where two of the four values (00b and 01b) report operational values, such as “on” and “off,” one value (10b) indicates data error, and one value (11b) indicates parameter not available or unsupported.

5.2.9 Parameter Ranges

For SAE J1939 defined parameters, the range of possible encoded values may be partitioned into one or more ranges: a range for valid parameter data, a range for parameter data error indicators, a range for parameter support indicators, and ranges for parameter-specific/future data use.

The encoded value partitioning for a parameter (SP) is derived from the SLOT specified for the parameter. The number of partitioned ranges and the value size of the partitioned ranges are a function of the data bit length and the parameter type.

Encoded values are the data presentation form used for communicating parameter data in the PG data. The SLOT defined for a parameter (SP) specifies the transfer function for converting between the engineering data value and the encoded data value.

The "Valid Signal" value range in Table 2 defines the range of encoded (logical) value for communicating numeric value data (i.e., data where a linear equation is used to translate between engineering value and the encoded value). The first two entries in Table 3 show the "Valid Signal" value range used for "measured" state value (discrete) data, and the first two entries in Table 4 show the "Valid Signal" value range for "status" state value (control) data.

The "Error Indicator" value range defines the range of encoded values used to indicate valid parametric data is not provided due to some type of error in the sensor, subsystem, or module. If a component failure prevents the transmission of valid data for a parameter, an "Error Indicator" value, described in Tables 2 and 3, should be used in the PG data in place of that parameter's data. However, if the measured or calculated data has yielded a value that is valid yet exceeds the defined parameter range, an "Error Indicator" value shall not be used; instead, encoded value for the appropriate minimum or maximum parameter value shall communicated.

The "Not Available" value range defines the range of encoded values indicate valid data for a parameter is not provided but not because of an error condition. In Table 2, the range is shown in the column titled "Not Available or Not Requested." In Table 3, the range is shown with the label "Not available or not installed." In Table 4, the range is shown with the label "Don't care/Take no action." The terms used in SP definitions in reference to this range can be these labels or similar. "Not Available" describes the use of these encoded value(s) in PG data for the parameter when the data provider cannot provide valid parameter data (e.g., due to current operating conditions) or the data provider does not support the parameter. SAE J1939 does not provide a means to explicitly indicate the reason the parameter data is not available. "Not Requested" and "Don't care/Take no action" describe the use of these encoded value(s) in PG data for the parameter to indicate when valid data is not provided for a command or action request parameter when data provider is not directly requesting action or data provider is not commanding a change.

The "Parameter-Specific Indicator" value range defines the range of encoded values available for valid parameter data using a data encoding that differs from how data is encoded with the "Valid Signal" encoded values. For example, a parameter can use the "Valid Signal" value range to communicate scaled numeric data (such as current gear) and use the "Parameter-Specific Indicator" value range to communicate state data (such as vehicle is in park). The meaning of these encoded values is specific to each SP. When parameter-specific indicator values are used by an SP, the SP definition shall explicitly define the data described by each encoded value.

The "Operational Range" for a parameter describes the range of encoded values consisting of the "Valid Signal" range and any "Parameter-Specific Indicator" values defined for the parameter.

Some parameters use the full set of encoded values for valid parameter data and, therefore, have no means to indicate a data error or not available. This is generally limited to SLOTS used for text data parameters, counters data that wraps around once the limit is achieved, CRC calculations, manufacturer ID, a Parameter Group Number (PGN), a Suspect Parameter Number (SPN), failure mode indicator (FMI), and some bit-mapped parameters.

Table 2 - Logical signal ranges

Parameter Size	Valid Signal	Parameter-Specific Indicator	Reserved Range for Future Indicator Bits
4 bits	0h to Ah	Bh	Ch to Dh
8 bits	0h to FAh	FBh	FCh to FDh
10 bits	0h to 3FAh	3FBh	3FCh to 3FDh
12 bits	0h to FAFh	FB0h to FBFh	FC0h to FDFh
16 bits	0h to FAFFh	FB00h to FBFFh	FC00h to FDFFh
20 bits	0h to FAFFFh	FB000h to FBFFFh	FC000h to FDFFFh
24 bits	0h to FAFFFFh	FB0000h to FBFFFFh	FC0000h to FDFFFFh
28 bits	0h to FAFFFFFFh	FB00000h to FBFFFFFFh	FC00000h to FDFFFFFFh
32 bits	0h to FAFFFFFFFh	FB000000h to FBFFFFFFFh	FC000000h to FDFFFFFFFh

Parameter Size	Error Indicator	Not Available or Not Requested
4 bits	Eh	Fh
8 bits	FEh	FFh
10 bits	3FEh	3FFh
12 bits	FE0h to FEFh	FF0h to FFFh
16 bits	FE00h to FEFFh	FF00h to FFFFh
20 bits	FE000h to FEFFFh	FF000h to FFFFFh
24 bits	FE0000h to FEFFFFh	FF0000h to FFFFFFFh
28 bits	FE00000h to FEFFFFFFh	FF00000h to FFFFFFFFh
32 bits	FE000000h to FEFFFFFFFh	FF000000h to FFFFFFFFh

Table 3 - Transmitted values for 2-bit discrete parameters (measured)

Range Name	Transmitted Value
Disabled (off, passive, etc.)	00b
Enabled (on, active, etc.)	01b
Error indicator	10b
Not available or not installed	11b

Table 4 - Transmitted values for 2-bit control commands (status)

Range Name	Transmitted Value
Command to disable function (turn off, etc.)	00b
Command to enable function (turn on, etc.)	01b
Reserved	10b
Don't care/take no action (leave function as is)	11b

5.3 SLOT

A SLOT is a specification for encoding parameter data. SLOT is an acronym for Scaling, Length, Offset, Transfer function. A SLOT definition specifies the bit length for encoded data, scaling factor, conversion offset value, engineering data units, and range of valid engineering values. SLOTS are used to manage and minimize the number of different data encodings used by SAE J1939 data parameters. When assigning a new SAE J1939 SP, SAE J1939 shall use an existing SLOT when the bit length, scaling factor, units, and engineering data range satisfy the needs for the parameter. The “Diagnostic” SLOT has no data encoding definition and is only used with diagnostic-only SPs.

5.3.1 SLOT Name

The SLOT Name is a unique textual identifier assigned to a SLOT. SAE J1939 SP definitions can use the SLOT Name to cross-reference the SLOT used for encoding the parameter data.

5.3.2 SLOT Identifier

The SLOT Identifier (SLOT ID) is a unique numeric identifier assigned to a SLOT. SAE J1939 SP definitions can use the SLOT ID to cross-reference the SLOT used for encoding the parameter data.

5.3.3 SLOT Type

The SLOT Type describes the physical quantity, property, or data type relationship of the SLOT.

5.3.4 Transfer Function (Data Encoding)

Transfer Function Type describes the model used for transforming between engineering (real world) values and encoded values. There are five main types of data encoding or transfer functions: scaled data (numeric), text, state value, bit mapped, and byte array.

5.3.4.1 Scaled Data (Numeric)

For numeric or scaled data, SAE J1939 uses a linear equation model for converting between engineering data and the encoded (transmitted) data. A scaled data SLOT specifies the bit length of the encoded data, the bit scaling factor, offset value, and engineering data units for converting between engineering data and the encoded data. The conversion between engineering data and encoded data is:

$$[\text{engineering value}] [\text{unit}] = ([\text{encoded value}] * [\text{scale factor}]) + [\text{offset}]$$

The SLOT scale factor only specifies the finest resolution of engineering data that can be communicated by encoding; the scale factor does not impose a requirement on the measurement capability of the reporting device.

5.3.4.2 Text Data

Text data describes encoding where the data is an ordered series of zero or more graphic characters and control function codes. See 5.4 for more details about text data SLOTS and parameters. The misnomer “ASCII” is used sometimes in the SAE J1939 documents when referring to text data and text SLOTS.

5.3.4.3 State Value Data

State value data, sometimes called value list data, describes encoding where an engineering (real world) meaning is assigned to (or communicated by) a specific encoded value. An SP using state value data encoding defines the mappings between encoded values and engineering (real world) meanings.

5.3.4.4 Bitmapped Data

Bitmapped data describes encoding where the encoded value is treated as a collection of bits, where each bit is assigned to identify or describe some attribute or piece of data. The encoded value can communicate multiple different, and possibly mutually exclusive, attributes or pieces of data.

5.3.4.5 Byte Array Data

Byte array data describes data encoding where the data is a sequence of one or more whole bytes of binary data treated as an array of bytes and the meaning of each data byte is parameter or manufacturer specific. For example, a data security certificate. Unless otherwise specified, each encoded byte can be a value from 0 to 255, inclusively.

5.3.5 Scaling (Scale Factor)

For a SLOT with scaled data encoding, the scaling property specifies the scalar multiplier, in engineering units, for each encoded value. For a scaled data SLOT, the scaling shall be a positive, non-zero value.

When defining a new scaled data SLOT, the scaling for the new SLOT should be an integral power of two of the scaling of another SLOT in the same SLOT type.

5.3.6 Offset

For a SLOT with scaled data encoding, the offset property specifies the smallest value, in engineering units, that can be communicated by the data encoding. For scaled data SLOTS, the offset is the engineering value communicated by the encoded value 0.

For scaled data SLOTS, the offset can be a negative value, zero, or a positive value.

- Specify a zero offset when the SLOT data range in only positive engineering values.
- Specify a positive offset value when the SLOT data range is only positive engineering values and does not need to encode the engineering value 0.
- Specify a negative offset value when the SLOT data range includes negative engineering values.
- When defining a SLOT with a negative offset value, the offset value shall be an integral multiple of the SLOT scaling to guarantee there is an encoded value for the engineering value 0.
- When defining a SLOT with a negative offset value, the offset should be specified using one of the following options:
 - a. Set offset to negative value that results in equal range of negative and positive engineering values around the 0 engineering value, e.g., offset of -125 yielding engineering data range of -125 to 125.
 - b. Set offset to negative values yielding unequal range of negative and positive engineering values around the 0 engineering value. This option should be used only when the physical quantity has absolute minimum engineering value, such as -273 °C for temperature SLOTS.

5.3.7 Length (Data Length)

The data length for the encoded parameter data for the SLOT. Length is expressed in units of bits or bytes, as appropriate. The data length can be specified as a fixed length or a variable length.

5.3.7.1 Length Minimum (Bits)

Numerical value describing the minimum number of bits which can be used for the encoded value (transmitted data) with the SLOT.

5.3.7.2 Length Maximum (Bits)

Numerical value describing the maximum number of bits which can be used for the encoded value (transmitted data) with the SLOT.

5.3.8 Unit

The unit property specifies the SI units for the engineering data.

Unless otherwise specified, all pressure SLOTS are measured as gage pressure.

5.3.9 Range (Data Range)

Describes the full range of values, in engineering units, that can be communicated for the data encoding based upon the defined data length, offset, and scaling. The range describes the smallest and largest values, in engineering units.

5.4 Text Data (ASCII) Parameters

This section describes the standard practices for text data parameters for SAE J1939 SPs. The misnomer “ASCII” is used sometimes in SAE J1939 documents when referring to text data and text SLOTS. SAE J1939 has three standard text SLOT models for different text length designation techniques, which are summarized in Table 5. Some of the text SLOT models use a delimiter technique for data length designation which reduces the permitted characters for parameter text data. The text SLOT models are discussed individually in more detail in 5.4.3, 5.4.4, and 5.4.5.

There can be multiple different text SLOTS defined by SAE J1939 for the same text SLOT model to accommodate different text lengths, delimiter characters, etc.

Table 5 - Summary of text SLOT models

Text SLOT Type	Description	Data Length Indication	Delimiter Character Included in Data Length?	Any Characters Not Allowed within the Data?	Required to Fill Data to a Specific Length?
Fixed Length Text	The data length is a fixed or constant number of bytes.	Fixed number of bytes of data.	No. The length indicates required data space for parameter data.	No. All of the text characters can be used in the data.	Yes. Shall provide data in all of the required number of bytes.
Character Delimited, Variable Length Text	The data length can vary within defined limits, and a specific text character is placed following the data to indicate the end of the parameter data.	Length indicated by the delimiter character (always required).	No. The length indicates allowed data space for parameter data.	Yes. The delimiter character is not allowed within the parameter data, since it will be interpreted as the delimiter.	No, unless the SP definition specifies a minimum data length of one or more characters.
Byte Count Delimited, Variable Length Text	The data length can vary within defined limits, and a separate data parameter (SP) within the PG data specifies the byte length of the text data.	Separate SP that specifies the ASCII data byte length.	No. The length indicates allowed data space for parameter data.	No. All of the text characters can be used in the data, unless specified otherwise in the SP definition.	No, unless the SP definition specifies a minimum data length of one or more characters.

5.4.1 Text Data Characters

Unless otherwise specified, text data uses an 8-bit single-byte coded graphic character encoding for 191 graphic characters according to the ISO 8859-1 Latin 1 Character set shown in Figure 9. The remaining 65 character values (0 through 31 and 127 through 159) are control codes and, according to ISO 8859-1, these character values are the C0 and C1 control codes as defined in ISO 6429. As specified in ISO 6429, the character value 0 (zero) is the “NULL” character code.

The terms “graphic text characters” and “graphic characters” is used in SAE J1939 to refer to the set of 191 character values defined by the ISO 8859-1 Latin 1 Character set. Though misnomers, the terms “ASCII characters” and “printable ASCII characters” are used sometimes in SAE J1939. The terms “non-graphic characters” and “text control codes” are used in SAE J1939 to refer to the set of 65 control code values defined as the C0 and C1 control codes as defined in ISO 6429. Though misnomers, the terms “ASCII control characters” and “non-printable ASCII characters” are used sometimes in SAE J1939.

Unless otherwise specified, only the graphic text characters are allowed in the data for text data parameters. The text control characters, or non-graphic text characters, are not allowed in the data for text data parameters, unless the parameter definition explicitly states otherwise. SPN 162 and SPN 163 are examples of text data parameters with explicit statements allowing the use of text control characters in the parameter data.

SLOT definitions and SP definitions can further allow or restrict the set of permitted characters. For example, an asterisk delimited text SLOT restricts the use of the asterisk – an asterisk cannot be used in the text except to delimit the end of the text.

Vertical boldface characters at the left in Figure 9 are the hexadecimal digit representing the upper nibble of the single byte code for the character. Horizontal boldface across the top characters in Figure 9 are the hexadecimal digit representing the lower nibble of the single byte code for the character. For example, the byte code for the capital letter “J” character is 4Ah.

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	----- should not be displayed -----															
1	----- should not be displayed -----															
2	space	!	“	#	\$	%	&	‘	(	)	*	+	,	-	.	/
3	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
4	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7	p	q	r	s	t	u	v	w	x	y	z	{		}	~	
8	----- should not be displayed -----															
9	----- should not be displayed -----															
A	nil	ı	ç	£	¤	¥		§	¨	©	ª	«	¬	-	®	—
B	°	±	²	³	´	µ	¶	·	¸	¹	º	»	¼	½	¾	¿
C	À	Á	Â	Ã	Ä	Å	Æ	Ç	È	É	Ê	Ë	Ì	Í	Î	Ï
D	Ð	Ñ	Ò	Ó	Ô	Õ	Ö	×	Ø	Ù	Ú	Û	Ü	Ý	Þ	ß
E	à	á	â	ã	ä	å	æ	ç	è	é	ê	ë	ì	í	î	ï
F	ð	ñ	ò	ó	ô	õ	ö	÷	ø	ù	ú	û	ü	ý	þ	ÿ

Figure 9 - ISO 8859-1 Latin 1 character set

5.4.2 Text Data Byte Order

The standard practice for the ordering of data bytes for text data parameters is defined in 5.1.9.

5.4.3 Fixed Length Text SLOT Model

The fixed length text SLOT model defines a text data encoding with a fixed, or non-varying, number of characters in the parameter data. All graphic text characters are available for use in the text data for an SP defined using this text SLOT model.

Some examples of SPs using a fixed length text SLOT are SPNs 162, 3620, and 4254.

5.4.3.1 SP Data Definition for Fixed Length Text

An SP defined using a fixed length text SLOT has the following data definition characteristics:

- The “Scaling” property indicates “Text” (some SAE J1939 documents can reference as “ASCII”)
- The “SP Length” property specifies a fixed byte length, such as “5 bytes”

The “SP Length” property defines the required byte length of the data for this SP. The SP data definition should specify the acceptable character(s) for an application to use to fill or pad the data to the specified length in case the text data is shorter than the required data length. The definition should indicate if there is a standard preference for inserting the pad or fill characters before or after the actual text data.

5.4.3.2 PG Data Details for Fixed Length Text Parameters

The PG data shall contain the specified number of bytes (SP Length) of data for the fixed length text data SP at the defined SP data position, whether the source application supports the SP or not.

If the source application supports the SP, then the source shall place the SP data of “SP length” bytes, as specified in 5.4.3.1, at the position defined for the SP in the PG.

If the source application does not support the SP, then the source application is still required to place “SP length” bytes of “not available” value (FFh) at the position defined for the SP in the PG.

5.4.4 Character Delimited, Variable Length Text SLOT Model

The character delimited, variable length text SLOT model defines a text data encoding with a varying number of characters followed by a specific character (delimiter) to indicate the end of the text data for the parameter. A graphic text character, such as an asterisk, or a control code character, such as NULL, can be specified as the delimiter. When a graphic text character is specified as the delimiter, then all graphic text characters except for the delimiter character are permitted in the data for an SP defined using with this text SLOT model; the delimiter character is not permitted in the SP data because it will be interpreted as the end of parameters text data. When a control code character is specified as the delimiter, then all graphic text characters are available for use in the data for an SP defined using this text SLOT model.

The delimiter character is not considered part of the text data for the parameter. Consequently, the delimiter character is not included in the maximum byte length value specified for the SLOT or SP. The delimiter is a mechanism within the PG data content to denote the end of the parameter data for the variable length text parameter. However, this fundamental perspective should not be seen as restricting how the parameter data is handled internally by an application.

Some examples of SPs using a character delimited, variable length text SLOT are SPNs 237 and 2902.

5.4.4.1 SP Data Definition for Character Delimited, Variable Length Text

An SP defined using a character delimited, variable length text SLOT has the following data definition characteristics:

- The “Scaling” property indicates “Text” (some SAE J1939 documents can reference as “ASCII”)
- The “SP Length” property specifies a variable length, such as “variable - up to 200 characters”
- The “SP Length” property specifies the delimiter character, such as “followed by an ‘*’ delimiter”
- The text data for the SP cannot contain the delimiter character

The “SP Length” property defines the maximum length for the text data for the SP, not counting the delimiter character. There is no minimum data length required for the data, unless the data length property explicitly states otherwise. The delimiter character is not included in the data length value specified in the “Data Length” property. The maximum data length is not a specification for the required number of bytes for the SP text data. A source application should not fill or pad the text data for an SP using this text SLOT model just to occupy the maximum length allowed.

The delimiter character is specified within the SLOT definition and SP definition because it places a restriction on the allowed text characters for the SP data. It is included in the “Data Length” property, since this property appears in the PG definition content.

The delimiter character is not considered part of the data for the parameter. As such, the SP data shall not include the delimiter character when the SP data is communicated other than as part of the PG data, such as through Memory Access Protocol with SP spatial addressing.

5.4.4.2 PG Data Details for Character Delimited, Variable Length Text

The PG data shall contain at least 1 byte and up to the defined maximum data length plus 1 number of bytes for the variable length text data, starting at the defined SP data position, regardless of if the source supports the SP or not. The designated delimiter character shall immediately follow the last text character for the SP data in the PG data. The delimiter character denotes the end of the SP data for the text data SP and indicates the starting position for any subsequent parameter. The delimiter is a mechanism within the PG data content to denote the end of the parameter data for the variable length text parameter.

If the source application supports the SP, then the source shall place the SP data followed by the delimiter, as specified in 5.4.4.1, at the position defined for the SP in the PG.

If the source application does not support the SP, then the source application is still required to place the delimiter character at the position defined for the SP in the PG.

The delimiter character is always required in the SP position for a delimited variable length text SP within the PG data. It is not necessary to include text data for a delimited variable length text parameter; however, the delimiter is always required. In other words, it is acceptable to transmit zero-length text for a variable length text parameter as long as the delimiter character is included in the PG data. The delimiter character is always required including situations when:

- The parameter is not supported by the source application
- The parameter is the only parameter in the PG data
- The parameter data is 0 bytes or characters in length
- The parameter is the last parameter in the PG data
- The parameter data length is the maximum data length allowed for the SP

The asterisk (*) character is the standard delimiter character for this text SLOT model. There are character delimited, variable length SLOTS that uses the NULL character as the delimiter character that are appropriate when there is a need to have the asterisk character available as a valid data character rather than a delimiter.

Several examples are provided below to illustrate the PG data content for several situations. For these examples, the letters “a” through “e” represent the data for five consecutive variable length text parameters (asterisk * delimited) within the PG data.

EXAMPLE 1: Data provided for each parameter. aaaa*bbb*c*dddd*eee*

EXAMPLE 2: Data only for parameters “a” and “b.” aaaaaaa*bbbbbbbbbb****

EXAMPLE 3: Data only for parameter “a” and “d.” *bbbbbbb**dddd**

EXAMPLE 4: Data only for parameter “e.” ****e*****

5.4.5 Byte Count Delimited, Variable Length Text SLOT Model

The byte count delimited, variable length text SLOT type model defines text encoding with a varying number of characters where a separate numeric parameter (SP) indicates the byte length of the text data. The text data SP and the associated text data length SP shall be transmitted in the same PG since the length of the text data can vary from one instance of the SP data to another instance of the SP data. All graphic text characters are available for use in the data for an SP using this text SLOT model.

Special design considerations shall be recognized by any application that is the source of a PG containing an SP using this text SLOT model. One design consideration involves maintaining synchronization between the text data length SP value and the length of the text data. Another design consideration involves the reported value for the text data length SP when the text data parameter is not available or supported by the source.

One example of SPs using a byte count delimited, variable length text SLOT are SPNs 509 and 3075, where SPN 509 is the text data SP and SPN 3070 reports the length of the text data for SPN 509. One example involves SPN 3075 as the text data SP and SPN 3072 reports the length of text data for SPN 3075.

5.4.5.1 SP Data Definition for Byte Count Delimited, Variable Length Text

An SP defined using a byte count delimited, variable length text SLOT has the following data definition characteristics:

- The “Scaling” property indicates “Text” (some SAE J1939 documents can reference as “ASCII”)
- The “SP Length” property indicates a variable length, such as “variable - up to 100 characters”
- The “SP Length” property does not specify a delimiter character
- The SP description shall identify the SP that reports the text data length

The “SP Length” property defines the maximum length available for the text data for the SP. There is no minimum data length required for the text data, unless explicitly stated otherwise. The maximum data length is not a specification of the required number of bytes for the SP text data. A source application should not fill or pad the text data for an SP using this text SLOT model just to occupy the maximum length allowed.

5.4.5.2 PG Data Details for Byte Count Delimited, Variable Length Text

The PG data shall contain 0 bytes up to the defined the maximum data length number of bytes for the variable length text data, starting at the defined SP data position, regardless of if the source supports the SP or not. The PG data also shall contain numeric data for the text data length SP at the defined SP data position for that SP, regardless of if the source supports the SP or not.

If the source application supports the SP, then the source shall place the text data, as specified in 5.4.5.1, in the PG data at the position defined for the text data SP and shall report the text data length in the PG data at the position defined for the associated text data length SP.

If the source application does not support the SP, then the source application shall report the data length value 0 (zero) as the data for the associated text data length SP.

The PG definition shall have the data text data length SP positioned in the PG data before the text data SP to enable a recipient application to determine the end of the text data. The source application shall make sure the value for the associated text data length SP is correct for the length of the text data.

The data for a subsequent parameter shall begin at the byte immediately follow the specified number of bytes from the starting byte position for the text data SP.

5.5 SP Position Notation and PG Data Placement

The placement of the SP data into the PG data is described by the combination of the SP Position (SP position in PG) specified for the parameter in the PG definition, the SP Length property of the SP definition, and occasionally on the data encoding (e.g., scaled, text, byte array). This section is intended to aid the reader in determining the proper placement of parameter data based upon the SP Position and SP Length properties. This section also serves as a guide for properly defining the “SP position in PG” property when assigning the placement of a new parameter to a PG.

For the purpose of parameter data placement discussion, the SP Positions are classified into one of several formats and parameter data lengths (SP Length) are classified into several types. Finally, parameter data placement is described for the different combinations of these two classifications. See 5.5.1 for the classifications of the SP Position notations and 5.5.2 for the parameter data length classifications. See 5.5.3 describes the diagrams in data placement illustrations provided for many combinations of the two classifications.

Generally, the “SP position in PG” notation reflects the bit position within the byte for the lowest order bit of the parameter data. When the parameter data is confined to a single byte, then the “SP position in PG” consists of one numerical value declaring the position for the lowest order bit of the parameter data. When the parameter data spans one or more byte boundaries, then the “SP position in PG” consists of two numerical values, each declaring the position for the lowest order bit of the parameter data in the lowest and highest order bytes. For numerical start position notation, the integer value identifies the byte, and the decimal value identifies the bit position (1 to 8, with 1 as the lowest order bit) within the byte. When the start position value does not include a decimal value, then the parameter data consumes the entire byte.

5.5.1 SP Position Notation Classifications for Data Placement

For the purposes of parameter placement discussion, the SP Position (SP position in PG) notations are classified based on the style used to describe position, as shown in Table 6.

Table 6 - SP position notation classifications

Classification	Description
Fixed	Describes a “SP position in PG” notation that specifies an absolute or fixed position for the parameter data in the PG data. Some examples of fixed start position notations are “3,” “5.4,” “1-2,” and “1.7-2.”
Equation	Describes a “SP position in PG” notation that specifies the position of the parameter data using an equation rather than an absolute position. Equation notations are appropriate when the parameter data length is variable, when the PG data has multiple variable length parameters, or when fixed length parameters follow variable length parameters in the PG data. Some examples of equation start position notations are “14-n,” “2 to n,” “5 to A,” and “A+1 to B.”
Field	Describes a “SP position in PG” notation that specifies the position of the parameter data in terms of its relative sequence in the PG data rather than with an absolute position or equation. Field start position notations are appropriate when the PG data has multiple consecutive variable length parameters or the parameter is repeated in the PG data. The placement order of fields follows the alphabetical sequence of the start positions. Some examples of field start position notations are “a,” “b,” and “c.”

5.5.2 Parameter Data Length Classifications for Data Placement

For the purposes of parameter placement discussion, parameter data lengths (SP Length) are classifications as shown in Table 7.

Table 7 - Parameter length classifications for SP position notation

Classification	Description
Integer Byte Length	Describes a parameter with a fixed data length that is an integer number of bytes. For example, a parameter with a data length of 1 byte, a parameter with a data length of 2 bytes, and a parameter with a data length of 8 bits.
Fractional Byte Length	Describes a parameter with a fixed data length that is not an integer number of bytes. For example, a parameter with a data length of 2 bits, a parameter with a data length of 5 bits, and a parameter with a data length of 10 bits.
Variable Byte Length	Describes a parameter with a variable data length that is an integer number of bytes. For example, a parameter with a data length of "variable - up to 200 characters." Text data and byte array parameters are the most common variable byte length parameters.

5.5.3 Parameter Data Placement Illustrations

Data placement illustrations for parameter placement notation styles help clarify the parameter placement practices (see Figures 10 through 18). These illustrations include one or more of the diagrams described in Table 8.

Table 8 - Diagrams in SP position notation illustrations

Diagram Title	Description
Data Definition	The data diagram serves to illustrate the parameter data bits for the example data, shown where the data bits go highest order bit to lowest order significant bit in a left to right manner. Individual bits are identified with a "b" followed by a number. The "b" is the abbreviation for "bit" and the number denotes the significance order of the bit, where bits with lower significance have a lower number value. In text data examples, the "b" identifier can be preceded by a "c" plus a number to designate the character instance.
PG Data Placement (Little-Endian)	The little-endian placement diagram illustrates the placement of the parameter data in the PG data as seen with little-endian structure. This diagram serves as a convenient way of discussing bit placement for the SAE J1939 data order practices since most parameter data is placed MSB first in the PG data.

5.5.4 Integer Byte Length Parameters - SP Position Notation and Data Placement

The information in Tables 9 and 10 presents the SP Position notations used with parameters with integer byte length data. The placement of the data bytes for integer byte length parameters larger than 1 byte depend upon the parameter data encoding (e.g., scaled, state value, text, byte array) and the corresponding data byte ordering (see 5.1.9). Examples of these SP Position notations and corresponding parameter data placement are illustrated in Figures 10 through 13.

When the parameter data length is 1 byte, the SP Position should consist of a single integer number identifying the PG data byte for the data. When the parameter data length is 2 bytes or more, the SP Position should consist of two integer numbers identifying the range of PG data bytes for the data.

Table 9 - SP position notation for integer byte length parameters (not text or byte array data)

SP Position	SP Length	Interpretation	Example Illustration
R	1 byte or 8 bits	Fixed position for a 1 byte data parameter where parameter data occupies the entire byte "R."	Figure 10
R-S R, S R (<i>obsolete</i>)	Y bytes or 16 bits	Fixed position for a multiple byte data parameter and the parameter data occupies "Y" number of bytes from byte "R" through byte "S." Since this parameter is not text or byte array data (based upon parameter definition), the data is positioned with the least significant byte in the lowest byte "R" and most significant byte in the highest byte "S," per 5.1.9. The SP Position should specify both the first and last bytes occupied by the parameter data.	Figures 11 and 12

Table 10 - SP position notation for integer byte length parameters (text or byte array data)

SP Position	SP Length	Interpretation	Example Illustration						
R	1 byte or 8 bits	Fixed position for a 1 byte data parameter where the parameter occupies the entire byte “R.”	Figure 10						
R-S	Y bytes	Fixed position for a multiple byte data parameter and the parameter occupies “Y” number of bytes from byte “R” through byte “S.” Since this parameter is text or byte array data (based upon parameter definition), the data is positioned with the most significant byte in the lowest byte “R,” per 5.1.9.	Figure 13						
“n”	Y bytes	Field position for a multiple byte data parameter and the parameter occupies “Y” number of bytes from the point that the field starts (i.e., in the first byte following the previous field). Since this parameter is text or byte array data (based upon parameter definition), the data is positioned with the most significant byte in the lowest byte, per 5.1.9. Example (PGN 64912, SPN 3560 and 3561) <table><tr><td>SP Position</td><td>SP Length</td></tr><tr><td>a</td><td>2 bytes (SPN 3560)</td></tr><tr><td>b</td><td>2 bytes (SPN 3561)</td></tr></table> <i>The structure of these two parameters repeats in the data field. The 2 bytes of data for SPN 3561 (field “b”) is placed in the 2 bytes following the last byte of SPN 3560 (field “a”).</i>	SP Position	SP Length	a	2 bytes (SPN 3560)	b	2 bytes (SPN 3561)	
SP Position	SP Length								
a	2 bytes (SPN 3560)								
b	2 bytes (SPN 3561)								

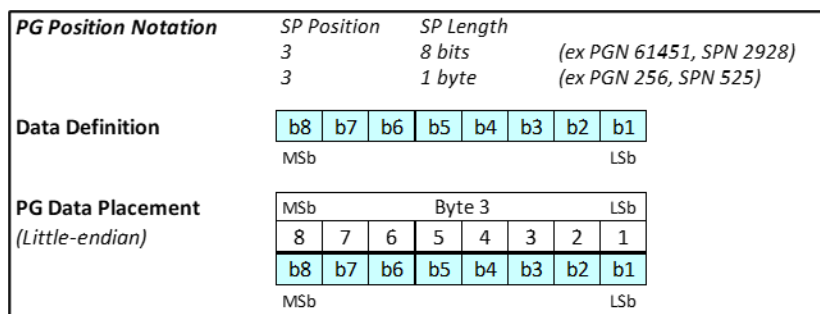


Figure 10 - Data placement for single byte data (all types of data)

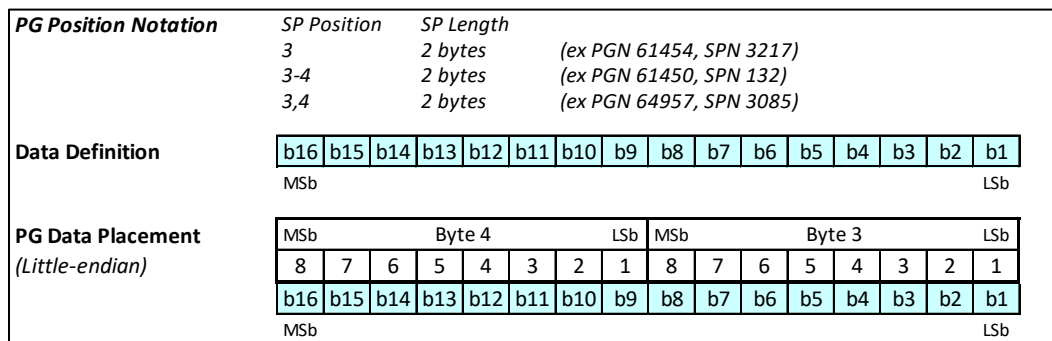


Figure 11 - Data placement for multiple byte data (not text or byte array data)

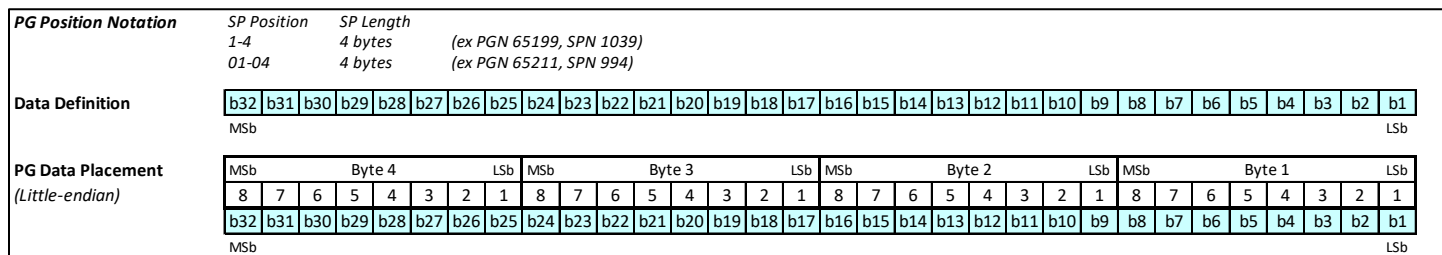


Figure 12 - Data placement for multiple byte data (not text or byte array data)

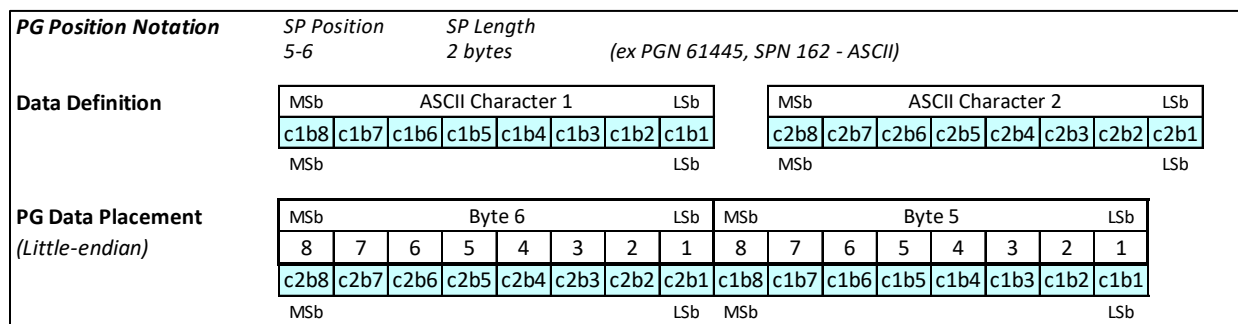


Figure 13 - Data placement for multiple byte data (text or byte array data)

5.5.5 Fractional Byte Length Parameters - SP Position Notation and Data Placement

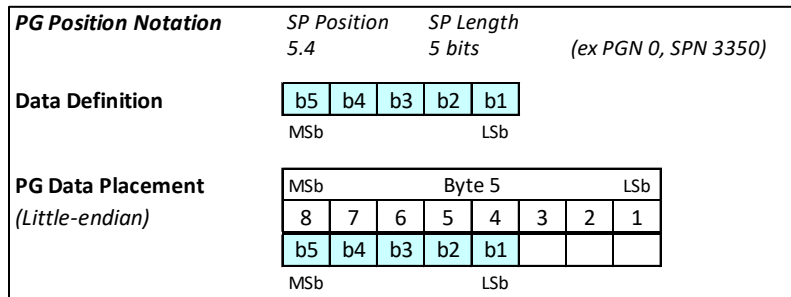
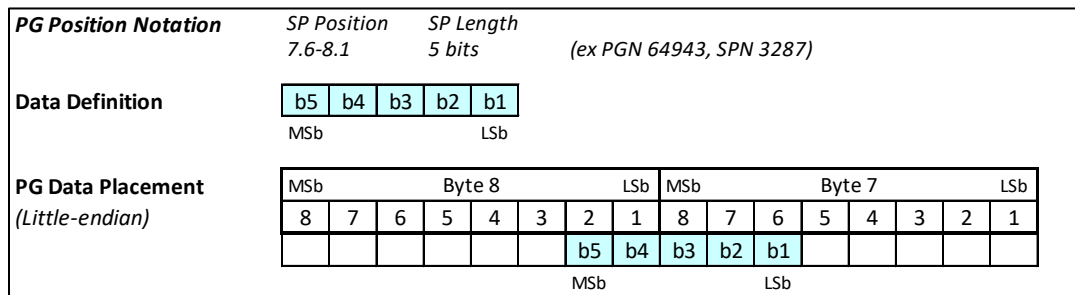
Fractional byte length parameters are parameters with a data length that is not an integer number of bytes, e.g., 2 bits, 5 bits, 10 bits, etc. The information in Table 11 presents the SP Position notations used with fractional byte length parameters and explains the respective parameter data placement. Figures 14 through 18 show examples of these SP Position notations and illustrate the parameter data placement.

The following guidelines explain how to determine data placement from the SP Position and SP Length properties for a parameter with fractional byte length data.

- a. In the SP Position notation, the number before the decimal point identifies the byte and the number after the decimal point identifies the bit position within that byte.
- b. If the parameter data length is less than 1 byte and all data bits are within the same byte, then the SP Position consists of one numerical value.
- c. If the parameter data length is larger than 1 byte or the data spans a byte boundary, then the SP Position consists of two numerical values separated by a comma or dash. The number before the comma or dash is the first position designation and the number after the comma or dash is the second position designation.
- d. If a position designation in the SP Position does not have a decimal value, then the start bit is at bit 1 in that byte. For example, a position designation of "2" is equivalent to the position designation "2.1." This abbreviated style is only used when the parameter data occupies the whole byte. In Table 11, a designation of "R" is equivalent to the designation "R.1," and a designation of "S" is equivalent to the designation "S.1." This is illustrated in Figures 16 through 18. In Figure 16, the second position designation is "2," so the lowest order data bit placed into byte 2 will be positioned at bit 1.
- e. For fractional byte length data, the least significant data bit is always positioned at the first position designation, and each next higher order data bit is placed into the next higher order data bit position. In Table 11, "R.x" represents the first position designation, so the least significant bit of the data is placed at bit "x" of byte "R," the next higher order bit of the data is placed at bit "x+1" of byte "R," etc. This is illustrated in Figures 14 through 18.
- f. When higher order data bit placement reaches a byte boundary and the next higher data byte is an intermediate byte between the bytes specified in the first and second position designations, then the next higher order data bit is placed at bit 1 of the next higher order data byte and additional higher order data bits are placed in next higher order fashion from that point. This is illustrated in Figure 18. In Figure 18, the SP Position notation identifies byte 6 in the first position designation and byte 8 in the second position designation, so byte 7 is an intermediate byte. When bit placement reaches byte 7, the next higher order data bit (bit "b9") is placed at bit 1 of byte 7, and the next higher order data bits are placed into byte 7 in next higher order fashion from that point.
- g. When higher order data bit placement reaches a byte boundary and the next higher data byte is the byte identified in the second position designation in the SP Position, then the number after the decimal in the second position designation indicates the bit position in that byte where the next higher order data bit is placed in the byte and any remaining higher order data bits are to be placed in next higher order fashion from that point. In Table 11, "S.w" represents the second position designation, so when data bit placement reaches byte "S" of the data, the next higher order bit of the data is placed at bit "w" of byte "S," the next higher order bit of data after that is placed at bit "w+1" of byte "S," etc. This is illustrated in Figures 15, 17, and 18. In Figure 18, the second position designation is "8.6." When bit placement gets to byte 8, then next higher order data bit, bit "b17," is placed at bit 6 of byte 8 and the last two bits, "b18" and "b19," are placed at bit 7 and bit 8 of byte 8, respectively.

Table 11 - SP position notation for fractional byte length parameters

SP Position	SP Length	Interpretation	Example Illustration
R.x	Y bits (Y less than 8)	Fixed position for a fractional byte length parameter less than 8 bits long and parameter data is within a byte boundary. The parameter data occupies "Y" number of bits of byte "R" with the least significant bit of the parameter data at bit "x" in byte "R" and the most significant bit of the parameter data is at bit ("x" + ("Y"-1)) in byte "R."	Figure 14
R.x-S.w (<i>obsolete</i>)	Y bits (Y less than 8)	Fixed position for a fractional byte length parameter less than 8 bits long and parameter data spans a byte boundary. The parameter data occupies the most significant bits of byte "R" from bit "x" to bit 8 and the remaining number of data bits start in byte "S" from bit 1 and end at bit "w." The least significant bit of the parameter data is placed at bit "x" in byte "R."	Figure 15
R.x-S	Y bits (Y greater than 8)	Fixed position of a fractional byte length parameter more than 8 bits long where the data crosses a byte boundary and stops on a whole byte. The parameter occupies the most significant bits of byte "R" from bit "x" to bit 8 plus all whole bytes up to "S."	Figure 16
R-S.w	Y bits (Y greater than 8)	Fixed position of a fractional byte length parameter more than 8 bits long where the data crosses a byte boundary and starts on a whole byte. The parameter occupies all whole bytes from "R" up to "S" and the remaining modulo-8 number of bits starting from bit "w" in byte "S."	Figures 17 and 18 (see Figure 18 ** Note)

**Figure 14 - Data placement for fractional byte data (less than 1 byte) within byte boundary****Figure 15 - Data placement for fractional byte data (less than 1 byte) across byte boundary**

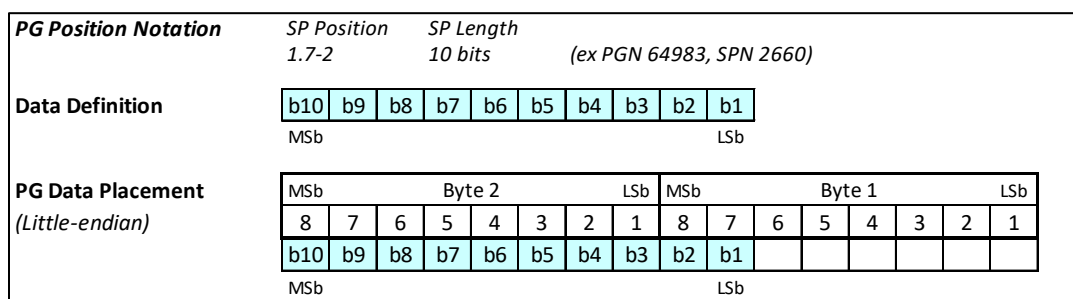


Figure 16 - Data placement for fractional byte data (larger than 1 byte) ending on byte boundary

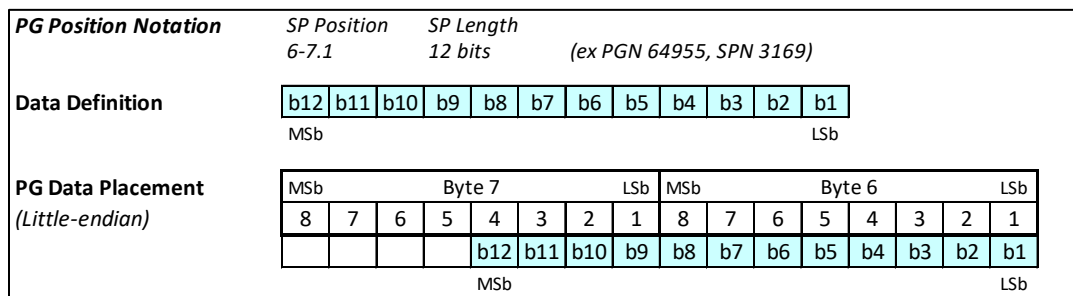


Figure 17 - Data placement for fractional byte data (larger than 1 byte) starting on byte boundary

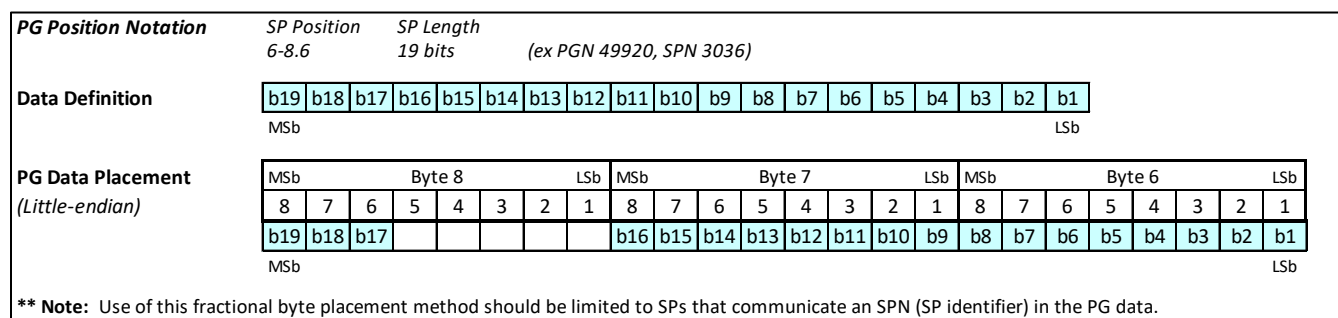


Figure 18 - Data placement for fractional byte data (larger than 1 byte) starting on byte boundary

5.5.6 Variable Length Parameters - SP Position Notation and Data Placement

The information in Table 12 presents SP Position notations used with variable length parameters. Text and byte array data parameters are the primary examples of variable length parameters. The SP Position is typically denoted using letters and equations to explain the position of the content within the PG data.

Table 12 - SP position notation for variable length parameters

SP Position	SP Length	Interpretation										
R-“N”	Variable - up to Y characters (“*” delimited)	The parameter data starts at byte “R” and continues through some variable number of bytes where the end is denoted by an asterisk character in the data stream. The length of the parameter data does not include the “*” delimiter. Example (PGN 65242, SPN 234) <table><tr><td>SP Position</td><td>SP Length</td></tr><tr><td>2-N</td><td>Variable - up to 200 characters (“*” delimited)</td></tr></table> <i>Parameter data starts at byte 2 and continues up to the asterisk character (at byte 203 at the highest).</i>	SP Position	SP Length	2-N	Variable - up to 200 characters (“*” delimited)						
SP Position	SP Length											
2-N	Variable - up to 200 characters (“*” delimited)											
R to “N” “N”+1 to “P”	Variable - up to Y characters Variable - up to Y characters	Equations define the starting position of two consecutive variable length parameters. The first parameter starts at byte “R” and continues through some variable number of bytes. The second parameter starts at the first byte after the last character of the first parameter and continues through some variable number of bytes. Example (PGN 64958, SPNs 3074 and 3075) <table><tr><td>SP Position</td><td>SP Length</td></tr><tr><td>2</td><td>1 byte (Number of bytes in SPN 3074 data)</td></tr><tr><td>3</td><td>1 byte (Number of bytes in SPN 3075 data)</td></tr><tr><td>5 to A</td><td>Variable - up to 100 characters (SPN 3074 data)</td></tr><tr><td>A+1 to B</td><td>Variable - up to 100 characters (SPN 3075 data)</td></tr></table> <i>First variable length parameter starts at byte 5 and continues through the number of bytes specified in byte 2. The second variable length parameter starts at first byte after SPN 3074 data and continues through the number of bytes specified in byte 3.</i>	SP Position	SP Length	2	1 byte (Number of bytes in SPN 3074 data)	3	1 byte (Number of bytes in SPN 3075 data)	5 to A	Variable - up to 100 characters (SPN 3074 data)	A+1 to B	Variable - up to 100 characters (SPN 3075 data)
SP Position	SP Length											
2	1 byte (Number of bytes in SPN 3074 data)											
3	1 byte (Number of bytes in SPN 3075 data)											
5 to A	Variable - up to 100 characters (SPN 3074 data)											
A+1 to B	Variable - up to 100 characters (SPN 3075 data)											
“n”	Variable - up to Y characters (“*” delimited)	Field position of a variable length data parameter. The parameter is the nth ordered field in the PG data. The parameter occupies the first data byte following the previous parameter data and continues some variable number of bytes where the end is denoted by an asterisk character in the data stream. The length of the parameter data does not include the “*” delimiter. Example (PGN 64965, SPN 2903) <table><tr><td>SP Position</td><td>SP Length</td></tr><tr><td>c</td><td>Variable - up to 200 characters (“*” delimited)</td></tr></table> <i>Parameter is the third field and continues up to the asterisk delimiter character (up to 201 bytes after the start of parameter data). The starting byte number depends upon the length of the data before this field.</i>	SP Position	SP Length	c	Variable - up to 200 characters (“*” delimited)						
SP Position	SP Length											
c	Variable - up to 200 characters (“*” delimited)											

5.6 Unspecified Bits in the PG Data Definition

Unspecified bits are the bits within the PG data that are not assigned to a parameter or are not used by the current parameters (SPs) in the PG definition. In SAE J1939 PG definitions, the unspecified bits are typically not shown or explicitly identified in the PG definition.

The “PG Data Length” property of the PG definition specifies the minimum and maximum byte length of the constructed PG data. The constructed PG data shall be at least the minimum length specified by the “PG Data Length” property, and all unspecified bits within the constructed PG data shall be filled with a value of one. This standard makes it possible to assign unspecified bits to parameters at some future time.

5.6.1 Unspecified Bits - Illustrated Example

An example of unspecified bits is provided in Figure 19 using a fictional PGN 12345. The top section of Figure 19 shows the PG data length and PG content definition for the PG. There are 36 unspecified bits in the PG definition in this example. The unspecified bits are bit 5 to bit 8 of byte 2 (4 bits total) and all bits in byte 5 through byte 8 (32 bits total).

The PG data definition indicates SPN_1 is a 1 byte parameter with a data position at byte 1. Since SPN_1 occupies all the bits in byte 1, there are no unspecified bits in byte 1. Next, the PG data definition indicates SPN_2 is a 4 bit parameter with an SP Position of “2.1,” which means the data for SPN_2 occupies bit 1 through bit 4 of byte 2. Since the next parameter has an SP Position starting in byte 3, this means bit 5 through bit 8 of byte 2 are Unspecified Bits. The third parameter in the PG data definition indicates SPN_3 is a 2 byte parameter with an SP Position of “3-4.” Since SPN_3 occupies all the bits in bytes 3 and 4, there are no unspecified bits in byte 3 or byte 4. Finally, the “PG Data Length” property indicates the PG data has fixed a length of 8 bytes, but the PG definition only lists parameter content through byte 4. All of the bits in byte 5 through byte 8 are Unspecified Bits. When the PG data is constructed, the PG data for this PG shall be 8 bytes in length, as specified by the “PG Data Length” property. The 36 Unspecified Bits shall be filled each with a one, and the other 28 bits for the data for SPNs SPN_1, SPN_2, and SPN_3 shall be filled appropriately.

PGN 12345 Example J1939 PG Definition

PG Data Length: 8 bytes

SP Position	SP Length	Parameter Name	SPN
1	1 byte	Example Parameter 1	SPN_1
2.1	4 bits	Example Parameter 2	SPN_2
3-4	2 bytes	Example Parameter 3	SPN_3

PG Data (Big-Endian)

Byte 1								LSb	Byte 2								LSb	Byte 3								LSb	Byte 4								LSb
8	7	6	5	4	3	2	1		8	7	6	5	4	3	2	1		8	7	6	5	4	3	2	1		8	7	6	5	4	3	2	1	
b8	b7	b6	b5	b4	b3	b2	b1		1	1	1	1	b4	b3	b2	b1		b8	b7	b6	b5	b4	b3	b2	b1		b16	b15	b14	b13	b12	b11	b10	b9	
SPN_1 data									Unspecified Bits				SPN_2 data					SPN_3 data																	

Byte 5								LSb	Byte 6								LSb	Byte 7								LSb	Byte 8								LSb
8	7	6	5	4	3	2	1		8	7	6	5	4	3	2	1		8	7	6	5	4	3	2	1		8	7	6	5	4	3	2	1	
1	1	1	1	1	1	1	1		1	1	1	1	1	1	1	1		1	1	1	1	1	1	1	1		1	1	1	1	1	1	1		
Unspecified Bits																																			

Figure 19 - Unspecified bits example

5.7 General Guidelines

5.7.1 Data Acquisition to Data Transmission Timing

It is the intent of the SAE J1939 network to provide current data and signals from a source so it can be used by other nodes. It is recommended that the time between physical data acquisition of data for an SP and the transmission of that data in a PG instance should be no more than two times the periodic transmission rate defined for the PG. Additional constraints can be defined for certain parameters (see 5.1.12.1).

5.7.2 Parameters with Multiple Sources

Each parameter received by a node for control purposes shall be configurable by the system integrator to identify the primary source of the data, as well as the secondary source, if applicable. It is to be expected that the system integrator configures each receiving device on a network identically. A secondary source of data is defined to be a device on the network that measures the data independently of the primary source of that data.

5.7.3 PG Data Length Variation

A controller application that receives a variable length or a fixed extendable PG (see 5.1.5.2) should be designed to accommodate the possibility of PG data length variation, including longer PG data, due to the assignment of new SPs to the PG data definition. Such a controller application should also accommodate the possibility of PG data length variation between different instances of the PG, whether from the same data provider or from different data providers.

The length of the PG data for an instance of a PG shall be no shorter than the PG Data Minimum Length (see 5.1.5.3) specified for the PG.

5.7.4 Retransmitting PG and SP Data

A device shall not receive SP data from the network segment and retransmit that same SP data using the same SP back onto the same network segment.

5.7.5 Assignment of SLOT for New Parameter (SP)

When a new SAE J1939 parameter is being defined, the SLOT assigned to the parameter defines the data encoding and encoded value range partitions for the parameter data, as discussed in 5.3 and 5.2.6. A new parameter should use an existing SLOT if one exists with scaling factor, range, and data length that meets or exceeds the new parameter data requirements. This permits maintaining data consistency, when possible, between parameters of a given SLOT type (temperature, pressure, speed, etc.). When necessary, a new SLOT may be defined with a different scaling factor, range, or data length. See 5.3 for guidelines on scaling and offset when defining a new SLOT.

5.7.6 Adding a Parameter to a Parameter Group

When a new data parameter is defined for SAE J1939, the parameter shall be assigned to a Parameter Group. A new parameter may be assigned to unassigned bits in an existing PG, appended to the end of an existing fixed extendable data length or variable data length PG, or assigned to a new PG. If existing Parameter Group definitions do not permit the inclusion of a new parameter, a new Parameter Group can be defined.

The following guidelines should be considered when grouping parameters together into an existing SAE J1939 PG or a new SAE J1939 PG. In most cases, grouping parameters based upon any one of these guidelines can end up violating one or more of the other parameter grouping guidelines. In general, parameters should be grouped into Parameter Groups as follows:

- a. By function (oil, coolant, fuel, etc.) and not by type (temperature, pressure, speed, etc.).
- b. With similar transmission rates (to minimize unnecessary overhead).
- c. By common subsystem (the device likely to measure and send data).
- d. Limit groups to 8 bytes of parameter data for PGs with transmission rates less than 200 ms. (This is to avoid Transport Protocol services when transmitted over an SAE J1939-21 data link.)

5.7.7 Naming Convention for Engine Parameters

When there are multiple instances of the same parameter on the same component (i.e., exhaust ports), the following naming convention will be used. While facing the engine flywheel housing, left bank (LB) parameters are assigned prior to the right bank (RB) parameters. Front parameters are assigned prior to the rear or back parameters (with the rear/back being the end containing the flywheel housing). For a six-cylinder in-line engine, the position furthest from the flywheel will be identified as 1, the position next closest to the flywheel will be identified as 2, etc. For a 12-cylinder "V" engine, the position furthest from the flywheel on the left bank will be identified as 1, the position furthest from the flywheel on the right bank will be identified as 2, followed by the position next closest to the flywheel on the left bank. Refer to SAE J824 and see Figure 20. When only one parameter is required or available, the parameter denoted as number 1 should be used. (i.e., an engine having only one turbocharger would use Turbocharger 1 Compressor Inlet Temperature when broadcasting the temperature).

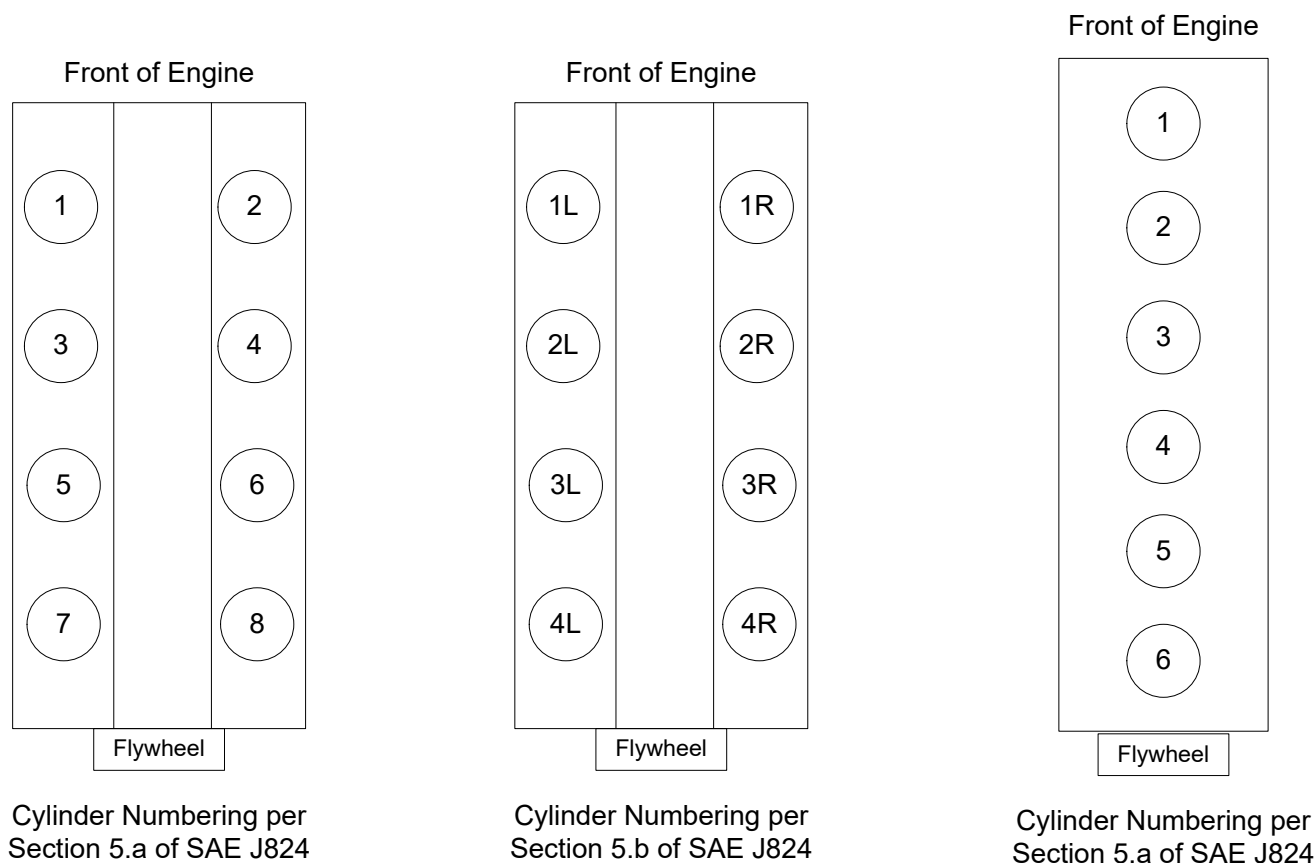


Figure 20 - Cylinder numbering per SAE J824

5.7.8 Signal and Parameter Terms

The terms “signal” and “parameter,” historically, have been used interchangeably, in most situations, to refer to a defined piece of data, i.e., a suspect parameter (SP). “Signal” is also used in some SP Labels and SP Descriptions to refer to the electrical signal to/from an electronic device, e.g., sensor, actuator, or switch. The SP Description shall provide clarification of the term’s usage when required.

6. NOTES

6.1 Revision Indicator

A change bar (|) located in the left margin is for the convenience of the user in locating areas where technical revisions, not editorial changes, have been made to the previous issue of this document. An (R) symbol to the left of the document title indicates a complete revision of the document, including technical revisions. Change bars and (R) are not used in original publications, nor in documents that contain editorial changes only.

APPENDIX A - SLOTS

Refer to SAE J1939DA for current listing of SLOTS.

APPENDIX B - SUSPECT PARAMETERS

Refer to SAE J1939DA for current listing of SPs.

APPENDIX C - PARAMETER GROUPS

Refer to SAE J1939DA for current listing of SPs mapped to PGs.

APPENDIX D - SUPPORTING INFORMATION

Refer to SAE J1939DA for supporting information and figures.